

# **BASIC PRINCIPLES OF MEDICINE 1:** **FOUNDATIONS TO MEDICINE**

## **LECTURE** **Signal Transduction**

**Dr. Shellon Thomas**  
**([sthoma10@sgu.edu](mailto:sthoma10@sgu.edu))**

**Acknowledgements: Sharmila Upadhyia, MBBS, MD,**  
**(Email: [shupadhyia@sgu.edu](mailto:shupadhyia@sgu.edu))**  
**Andrew Sobering, PhD**

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# Recommended reading

- Lippincott's Biochemistry 9<sup>th</sup> Edition   
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
  - Chapter 8: Introduction to Metabolism and Glycolysis) - Section: Metabolism Regulation
- [Lippincott® Illustrated Reviews: Cell and Molecular Biology, 3e | Premium Basic Sciences | Health Library](#); Chapters 17, 18, 19.
- Lippincott Illustrated Reviews: Biochemistry, 9e: Chapter 8 (Introduction to Metabolism and Glycolysis) - Metabolism Regulation: [Introduction to Metabolism and Glycolysis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Practice Questions: [8.6](#)
- Reference: Marks' Basic Medical Biochemistry: A Clinical Approach, 6e: Chapter 10. [Cell Structure and Signaling by Chemical Messengers | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library](#)

# Objectives: Signal Transduction

|                                 |                                                                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SOM.MK.I.BPM1.1.FTM.3.BCHM.0120 | Describe different types of cell signalling: endocrine, autocrine, paracrine, neuronal & contact-dependent.                                                                                                                 |
| SOM.MK.I.BPM1.1.FTM.3.BCHM.0121 | Indicate four different classes of receptors (steroid receptors, ion-channel receptor, receptor enzyme, receptors that form second messengers including G-protein-coupled receptors) with specific examples for each class. |
| SOM.MK.I.BPM1.1.FTM.3.BCHM.0122 | Discuss events involved in signal transduction                                                                                                                                                                              |
| SOM.MK.I.BPM1.1.FTM.3.BCHM.0123 | Identify the location of the receptors for steroid hormones. Describe intracellular signal transduction by steroid hormones.                                                                                                |
| SOM.MK.I.BPM1.1.FTM.3.BCHM.0124 | Describe the molecular changes when insulin binds to its receptor.                                                                                                                                                          |
| SOM.MK.I.BPM1.1.FTM.3.BCHM.0125 | Describe the different target (effector) enzymes of G proteins.                                                                                                                                                             |
| SOM.MK.I.BPM1.1.FTM.3.BCHM.0126 | Describe the two major second messenger systems (adenylate cyclase and phosphoinositide systems) associated with G-proteins and signal termination.                                                                         |

# Cell-Cell Signaling and Communication

- Key role in cellular processes such as proliferation, growth, differentiation, cellular maintenance and results in a variety of cellular responses
- Regulation of
  - Gene expression
  - Metabolism
  - Cell shape and cellular migration
  - Neuronal signaling
- Mis-regulation of signal transduction pathways
  - Cellular transformation (carcinogenesis)
  - Endocrine disorders like diabetes mellitus, thyroid disorders....

# Regulation of Metabolism

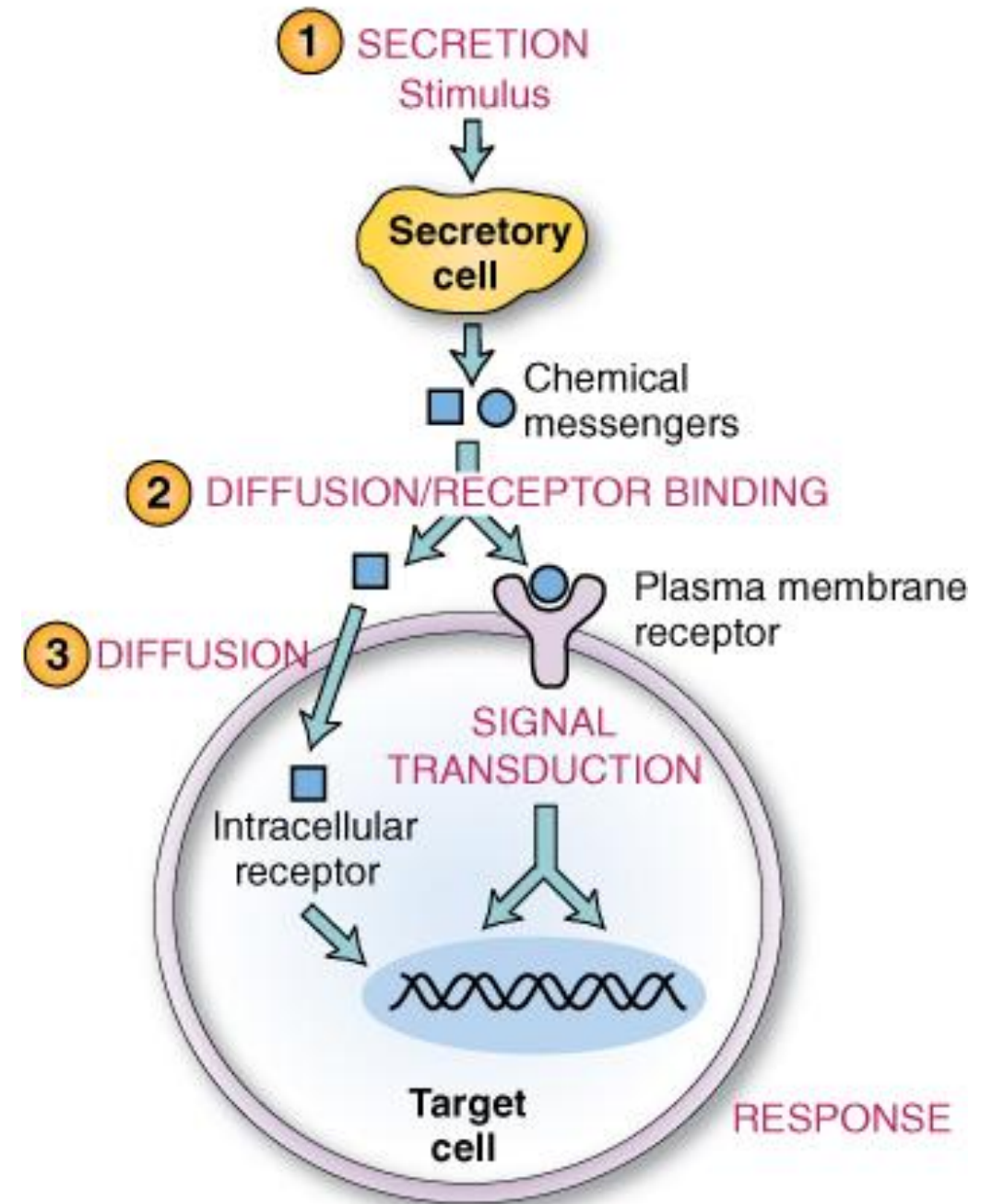
Metabolic pathways coordinated to produce energy or synthesize products to meet needs of cell and organism

Efficient ***communication system*** necessary to coordinate various functions

Regulation depends on:

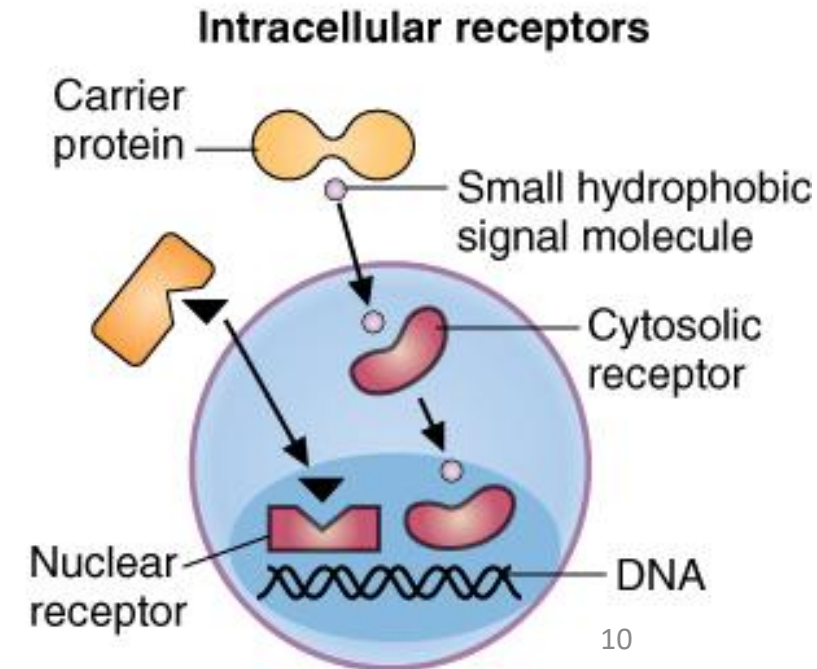
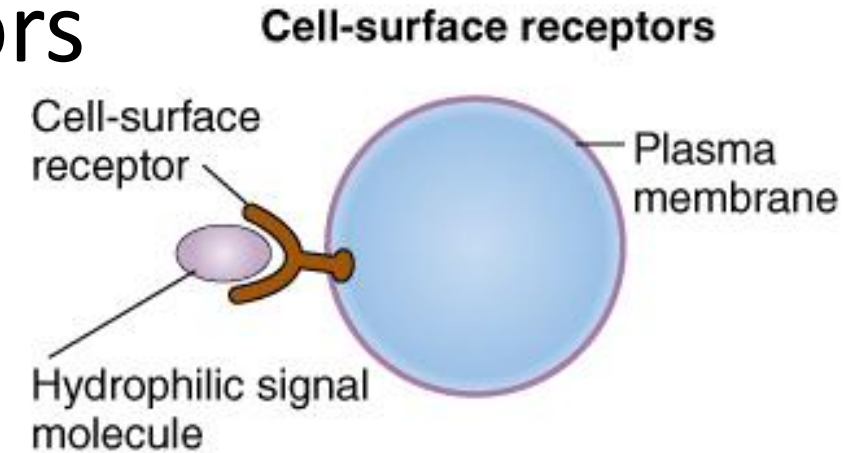
- ***Intercellular signals***
- ***Intracellular signals***

# Chemical (Intercellular) messengers



# Compare: Intracellular vs cell-surface (plasma membrane) receptors

- Plasma membrane receptors have extracellular binding domains for ligand binding
- Intracellular receptors bind lipophilic/hydrophobic (steroid hormones or other messengers) that diffuse through plasma membrane
- Intracellular receptors present in
  - Cytoplasm and translocate to the nucleus on binding to hormone, or
  - Nucleus and Hormone-receptor complex binds to DNA





# Intracellular vs cell-surface (plasma membrane) receptors

## Cell surface receptors

- Hormone hydrophilic
- Extracellular domain for binding hormone/ signal
- Intracellular domain for signal transduction
- Many produce second messengers within cells
- Rapid response; slower response effect on gene
- Examples: Peptide hormones – Insulin, glucagon, epinephrine; Acetylcholine; GABA; Eicosanoids

## Intracellular receptors

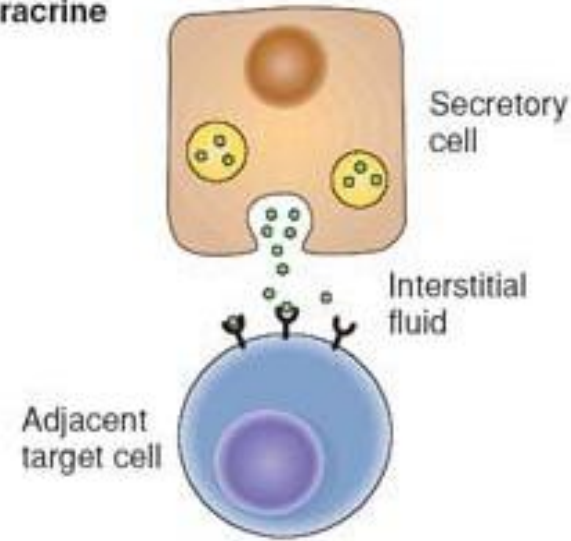
- Lipophilic/ hydrophobic signals diffuse through plasma membrane
- Located in nucleus or cytosol
- Conformational change on binding hormone
- H-R complex enter nucleus
- Bind to specific regions of DNA
- Influence transcription and translation (gene expression)
- Function like transcription factors
- Response is slower
- Steroid hormones, thyroxine, retinoic acid(Vitamin A) and vitamin D

## Intercellular signals

are converted to an intracellular signal in the adjacent cell

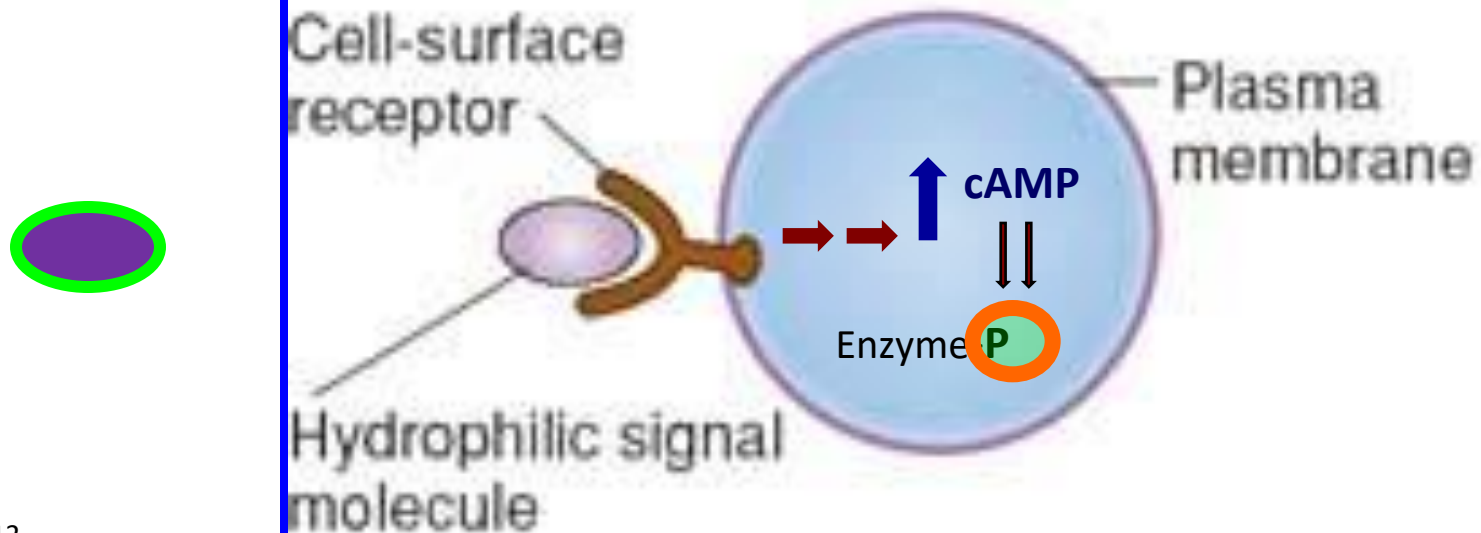
## Intracellular signals

Paracrine



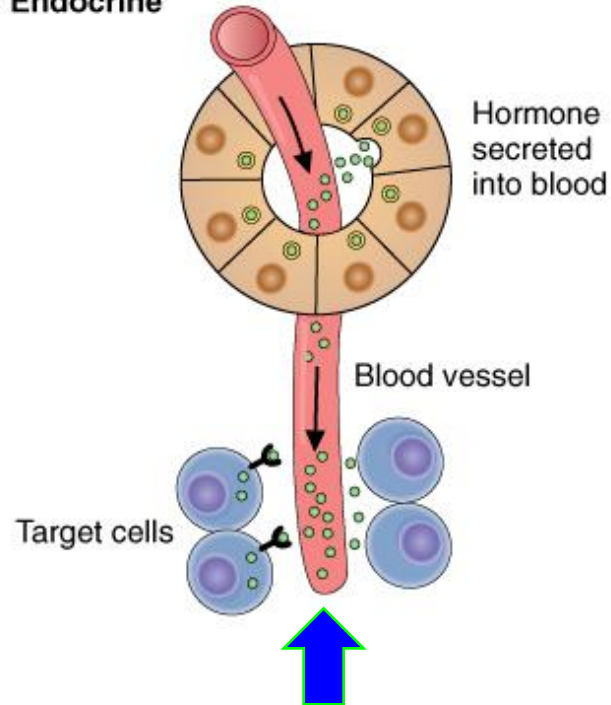
Many extracellular signals (hormones) exert their effects by producing intracellular second messengers

## Cell-surface receptors



## Endocrine

A. Endocrine



### Hormones:

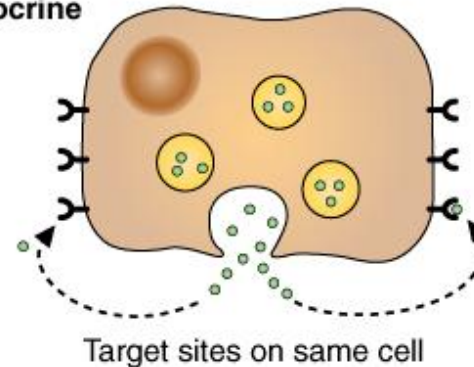
- Insulin
- Glucagon
- Epinephrine
- Steroid hormones
- Peptide hormones

# Types of intercellular signals

- Some growth factors
- Eicosanoids

## Autocrine

C. Autocrine

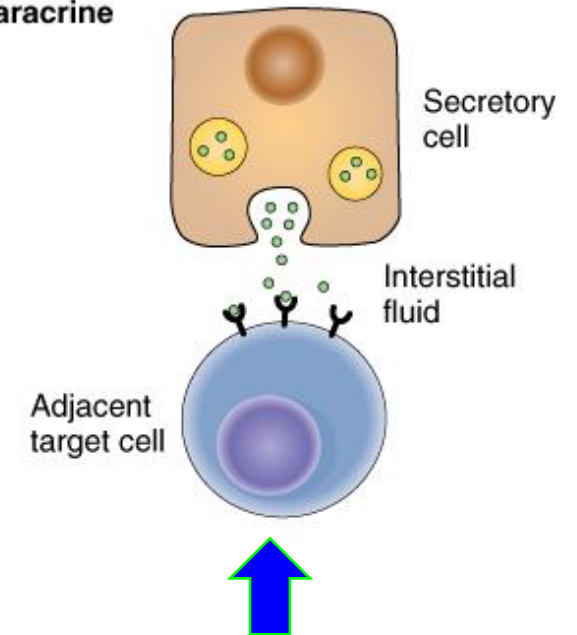


Y Receptor

• Hormone or other chemical messenger

## Paracrine

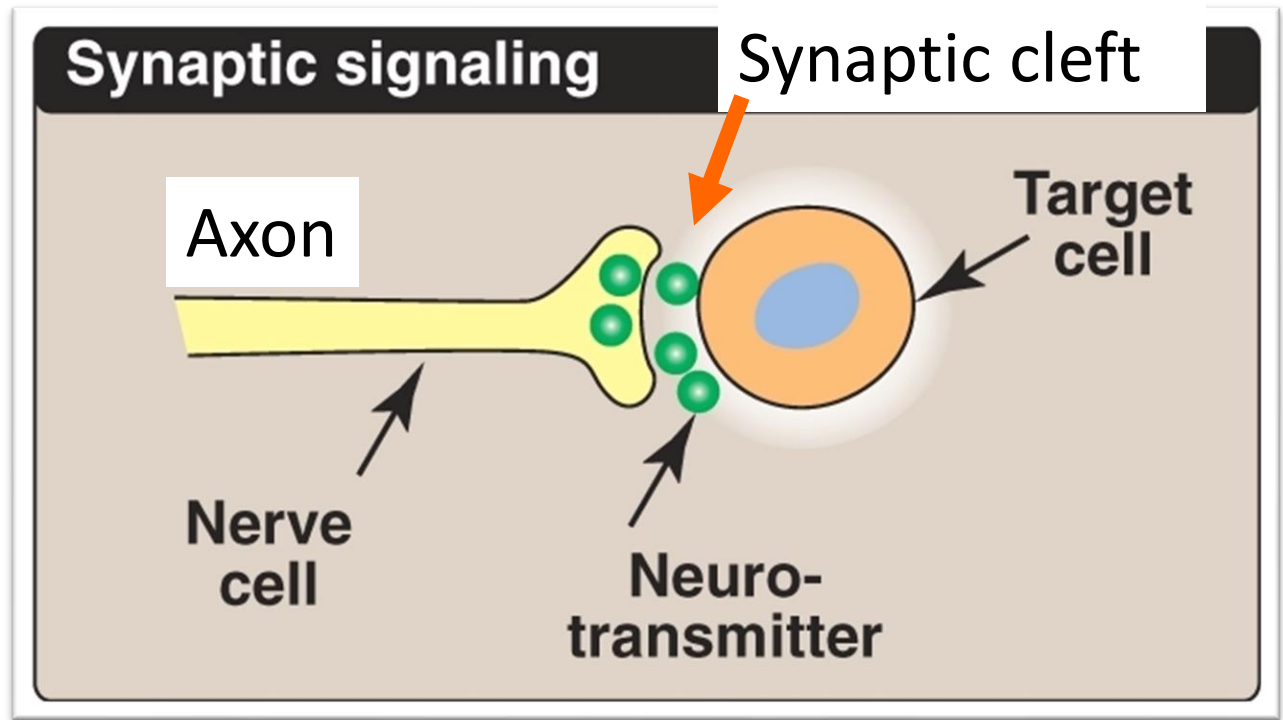
B. Paracrine



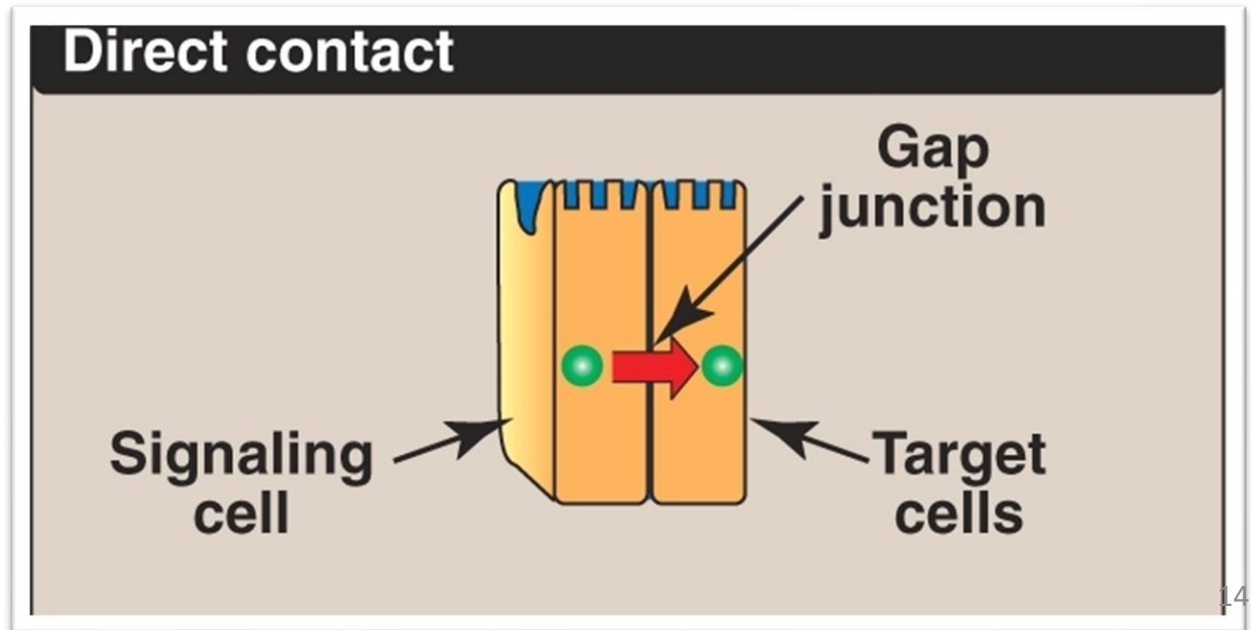
- Neurotransmitters
- Some Growth factors
- Eicosanoids
- Nitric oxide (gas)

# Other types of signaling

## Neuronal Signaling



## Contact Signaling



# Intracellular vs cell-surface (plasma membrane) receptors

## Cell surface receptors

- Hormone hydrophilic
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- Examples: Peptide hormones – Insulin, glucagon, epinephrine; Acetylcholine; GABA; Eicosanoids

## Intracellular receptors

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# Some Signals to Which Cells Respond

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Antigens

Cell surface glycoproteins/oligosaccharides

Developmental signals

Extracellular matrix components

Growth factors

Hormones

Light

Mechanical touch

Neurotransmitters

Odorants

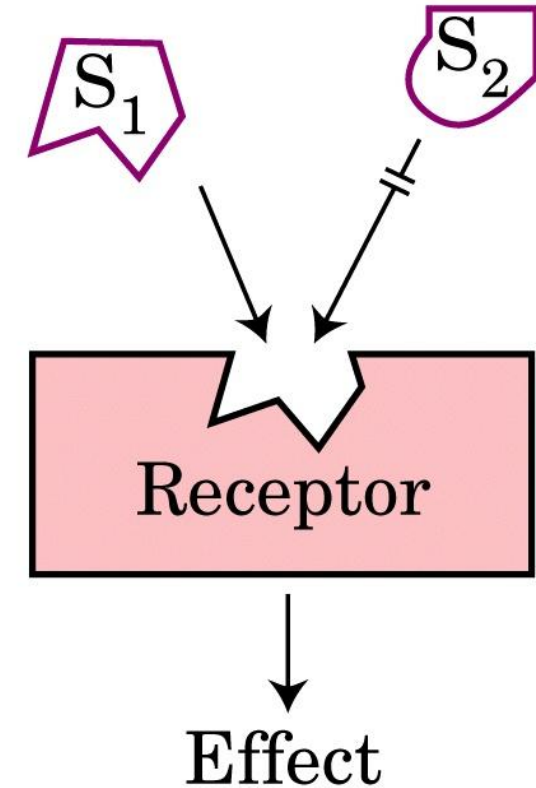
Pheromones

Tastants

# Features of intracellular signal-transduction (receptor-mediated) systems

## (a) **Specificity**

Signal molecule fits binding site on its complementary receptor; other signals do not fit.



Each hormone has specific receptors

A hormone may have more than one type of receptor:

**Epinephrine and Norepinephrine** have **alpha1, alpha2** and **beta** receptors (different tissues)

**Acetylcholine** has **nicotinic** and **muscarinic** receptors

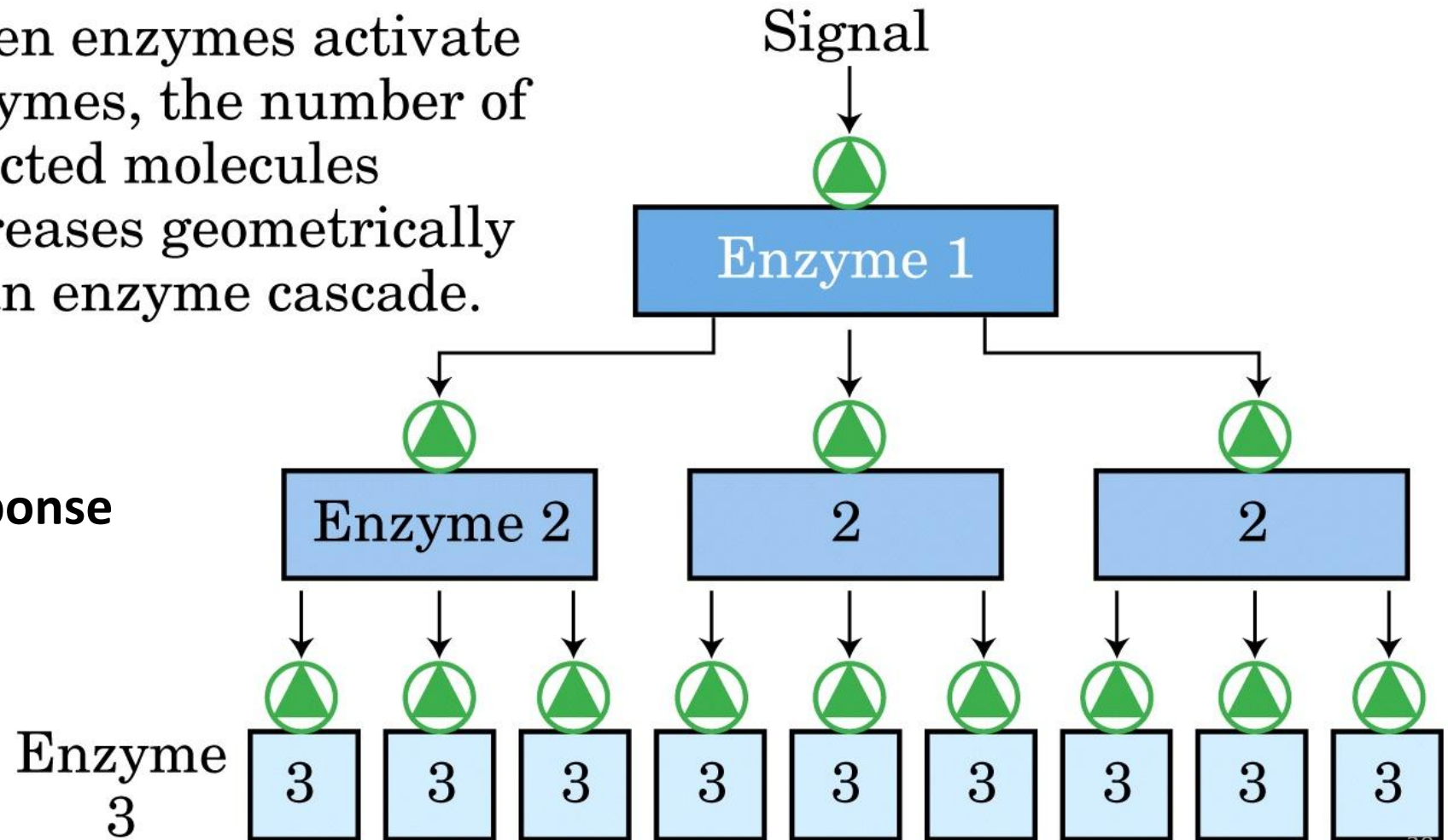


# Features of intracellular signal-transduction (receptor-mediated) systems

## (b) Amplification

When enzymes activate enzymes, the number of affected molecules increases geometrically in an enzyme cascade.

Rapid biological response

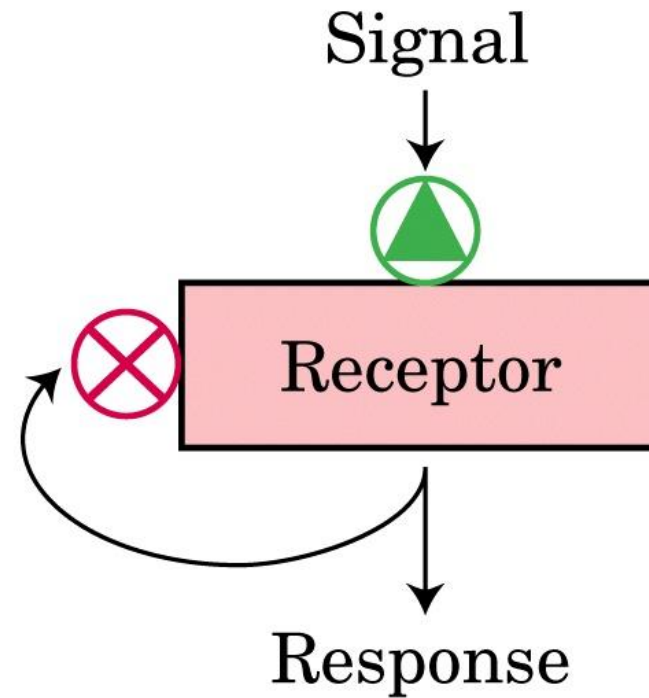




# Features of intracellular signal-transduction (receptor-mediated) systems

## (c) **Desensitization/Adaptation**

Receptor activation triggers a feedback circuit that shuts off the receptor or removes it from the cell surface.

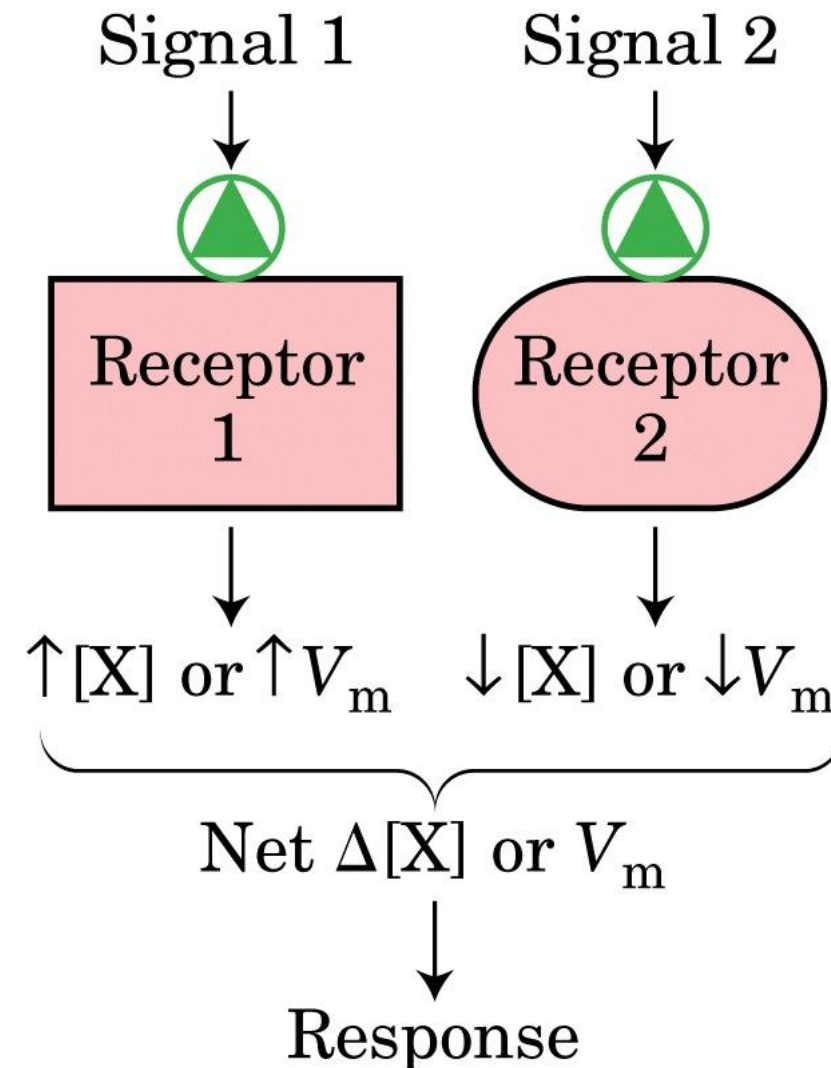


# Features of intracellular signal-transduction (receptor-mediated) systems

## (d) Integration

When two signals have opposite effects on a metabolic characteristic such as the concentration of a second messenger  $X$ , or the membrane potential  $V_m$ , the regulatory outcome results from the **integrated** input from both receptors.

**“Outcome” is additive**



# Features of intracellular signal-transduction (receptor-mediated) systems

## A. Specificity

- Specific signals (i.e. hormone) activate a specific receptor

## B. Amplification of signal

- One hormone molecule outside the cell can result in many thousands of activated molecules inside the cell

## C. Desensitization/Adaptation

- Must have a way to turn off the stuff that was activated, or to reduce responses to excessive signals

## D. Integration

- Many pathways feed into each other, output depends on network of all pathways

# Receptor-mediated Signal Transduction

(Extracellular signals producing *intracellular* signals)

4 basic types of **signal transduction pathways**  
mediated by **4 different types of receptors:**

1. Intracellular receptor (steroid receptor)
2. Gated-ion channel
3. Receptor enzyme (Catalytic receptor)
4. Receptors that form intracellular **second messengers**

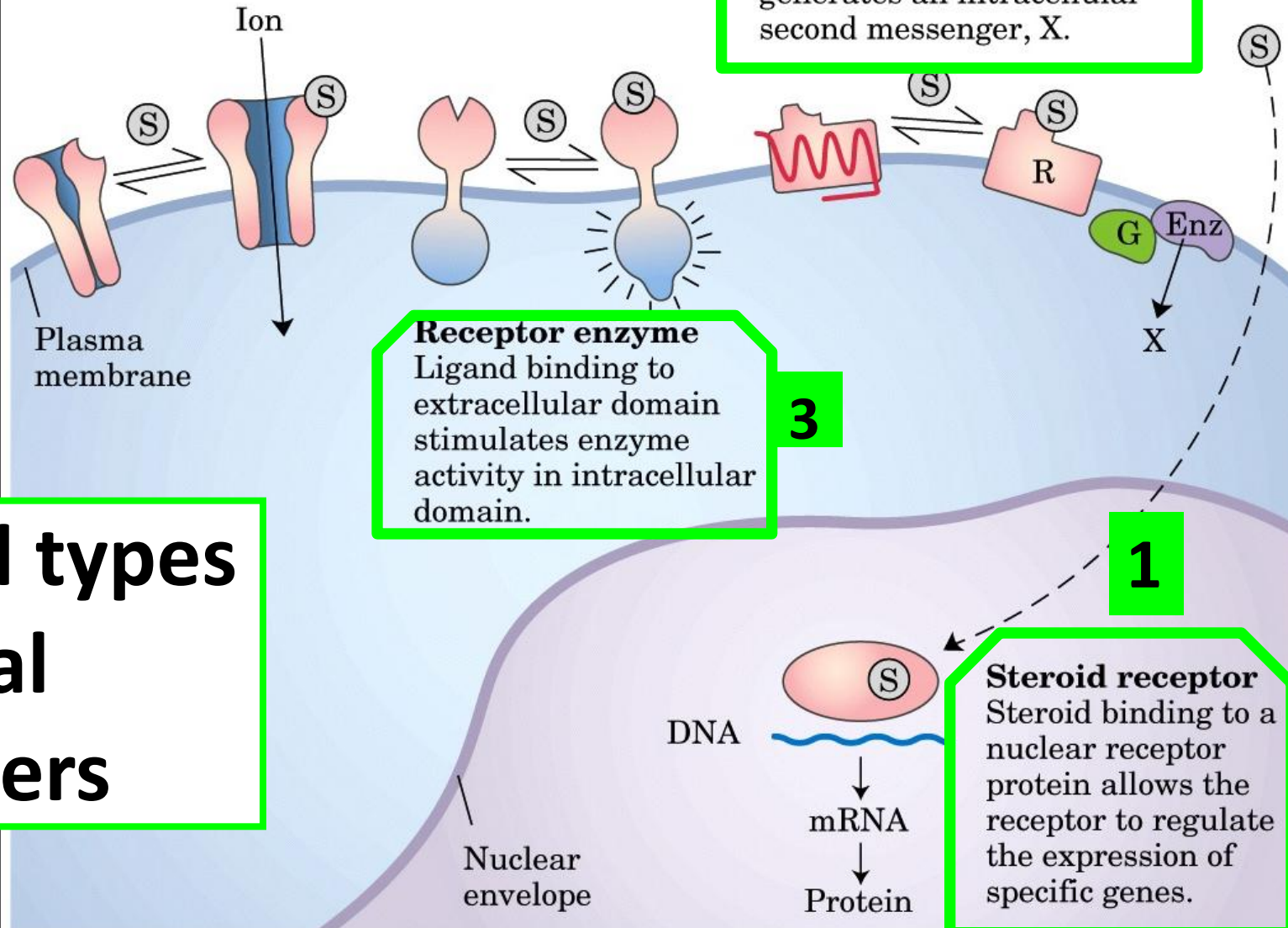
**Gated ion channel**  
Opens or closes in response to concentration of signal ligand (S) or membrane potential.

2

GPCR

**Serpentine receptor**  
External ligand binding to receptor (R) activates an intracellular GTP-binding protein (G), which regulates an enzyme (Enz) that generates an intracellular second messenger, X.

4



## Four general types of signal transducers

4 basic types of **signal transduction pathways**  
mediated by **4 different types of receptors**:

1. Intracellular (steroid) receptor
2. Gated-ion channel
3. Receptor enzyme (Catalytic receptor)
4. Receptors that form intracellular **second messengers**

# 1. Steroid and thyroid receptor family (Intracellular)

Lipophilic hormones cross cell membrane to bind ***specific intracellular receptors***

Examples include:

- Testosterone, estrogen, cortisol, aldosterone (steroid hormones)
- Vitamin D and retinoic acid (Vitamin A)
- Thyroid hormone (thyroid receptors)

Hydrophobic/ lipophilic hormones **induce (up-regulate) or repress (down-regulate)** gene expression at level of transcription



# Steroid-thyroid receptor family

Bind *intracellular* receptors

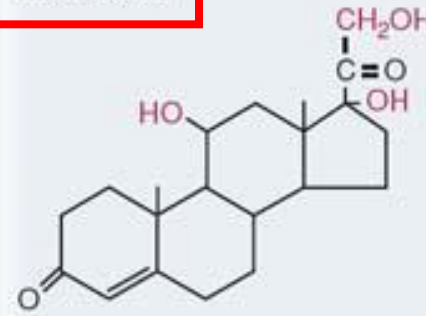


Change in expression of target genes

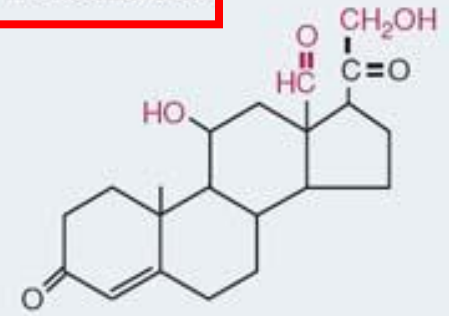


Increased production of protein (Induction) or Repression

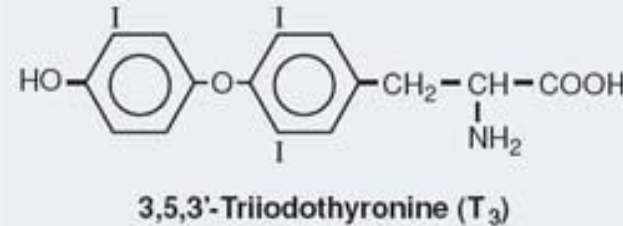
A. Cortisol



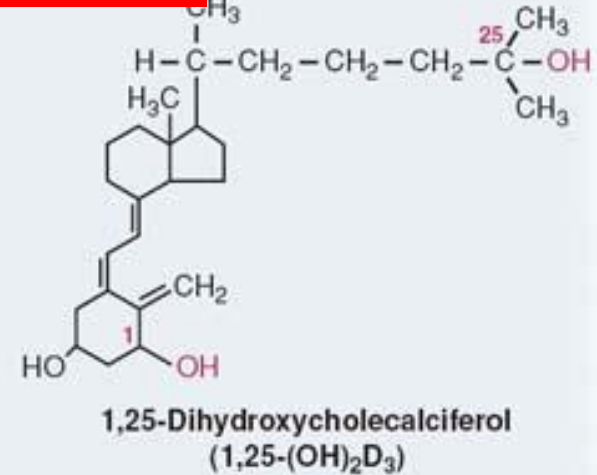
B. Aldosterone



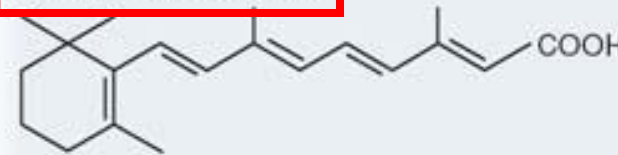
C. Thyroid hormone (T<sub>3</sub>)



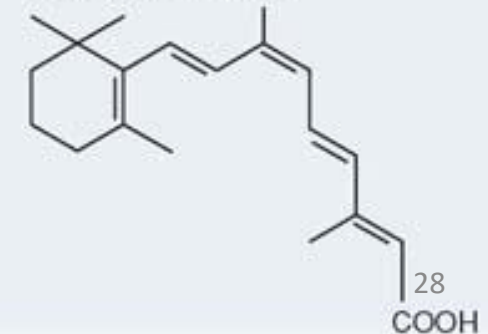
D. Vitamin D<sub>3</sub>



E. Retinoids  
All-trans retinoic acid



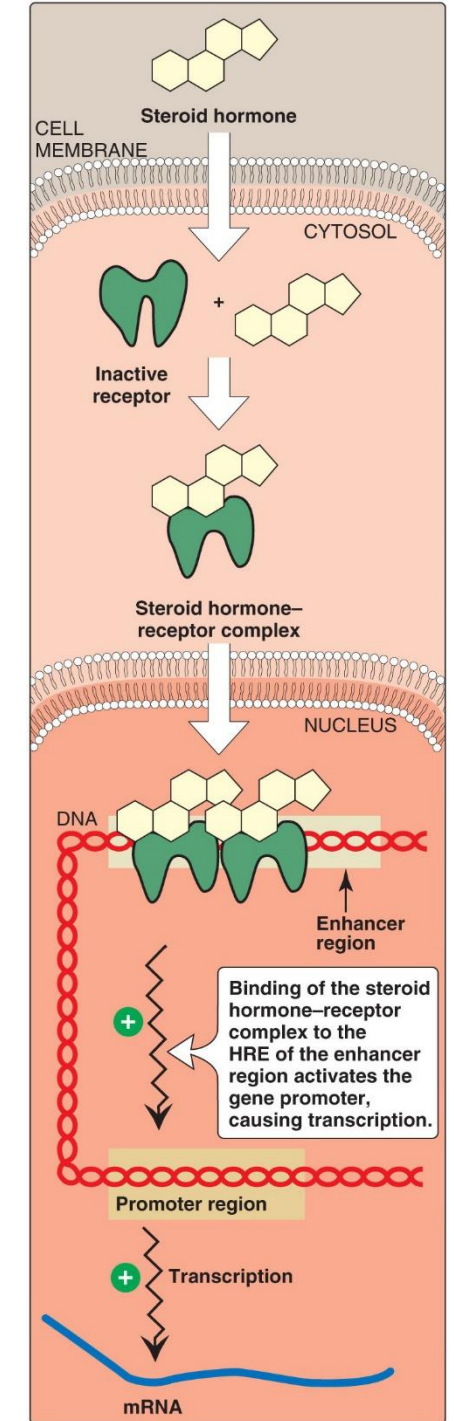
9-cis Retinoic acid





# Steroid receptor signal transduction

- Hormone diffuses into cell (lipophilic/ hydrophobic)
- Binds specific receptor (cytosol or nucleus)
  - Forms hormone-receptor complex; Conformational change in receptor
- H-R complex enters nucleus
- H-R complex binds specific regions of DNA (Hormone Response Elements; HRE)
- H-R complex binds enhancer regions, activates promoter, increases transcription (INDUCTION of gene expression) and translation to form protein → Cellular response
- If H-R complex binds repressor regions → decreased transcription (REPRESSION of gene expression)



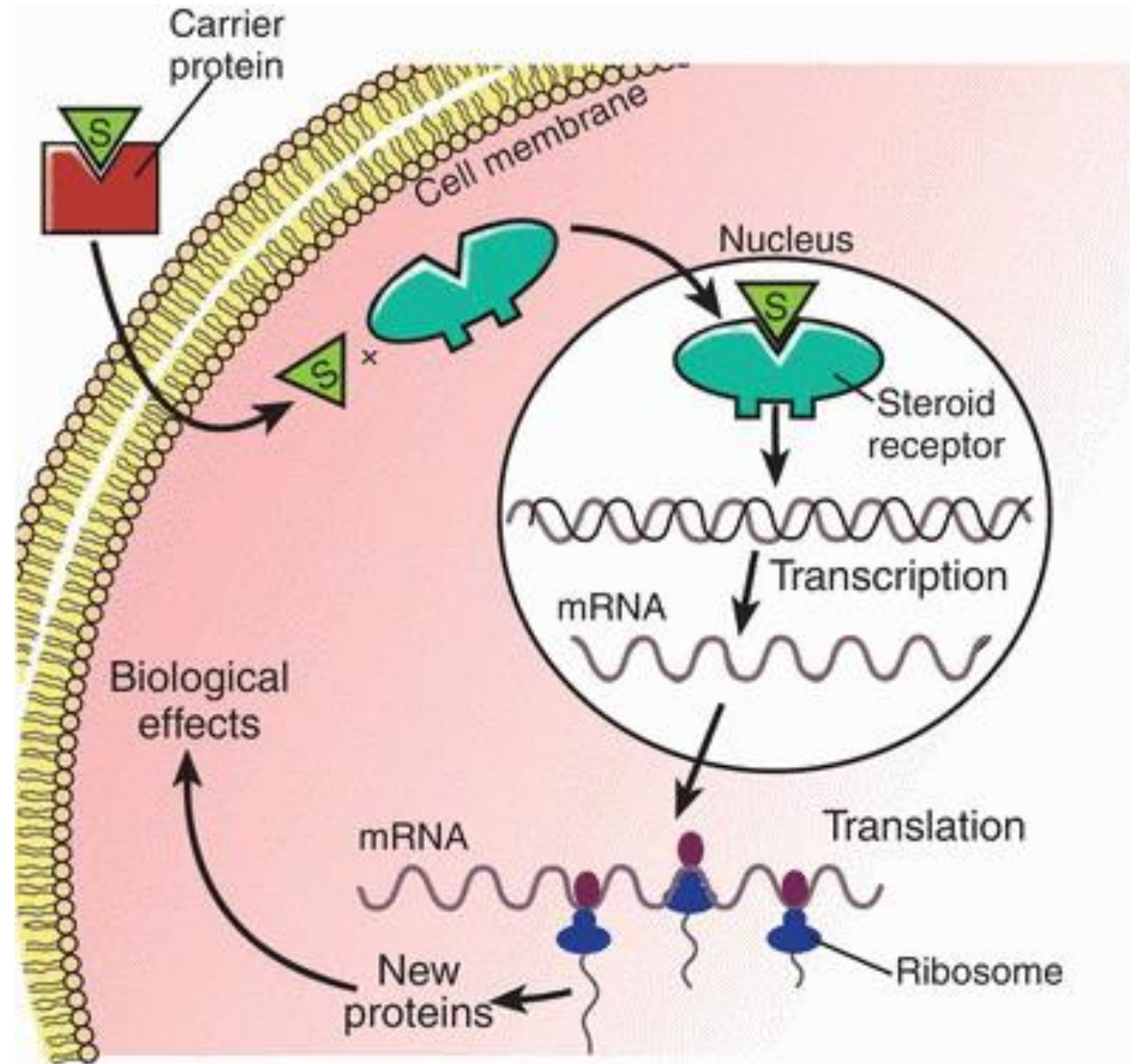
# Steroid receptor signal transduction

- Binding of Hormone-Receptor complex to HRE regulates expression of adjacent genes
- Increase or decrease rates of mRNA synthesis (Transcription)
- Cause INDUCTION or REPRESSION of gene expression
- Works through modulating gene expression
- Hormone-Receptor complex works like a transcription factor
- **Comparatively slow: response may take hours to days to be effective**

# Steroid receptor signal transduction

Works through gene expression

Comparatively slow;  
Response may take hours or days



4 types of **signal transduction pathways** mediated by  
**4 different types of receptors:**

1. Intracellular (steroid) receptor

2. Gated-ion channel

3. Receptor enzyme (Catalytic receptor)

4. Receptors that form intracellular **second messengers**

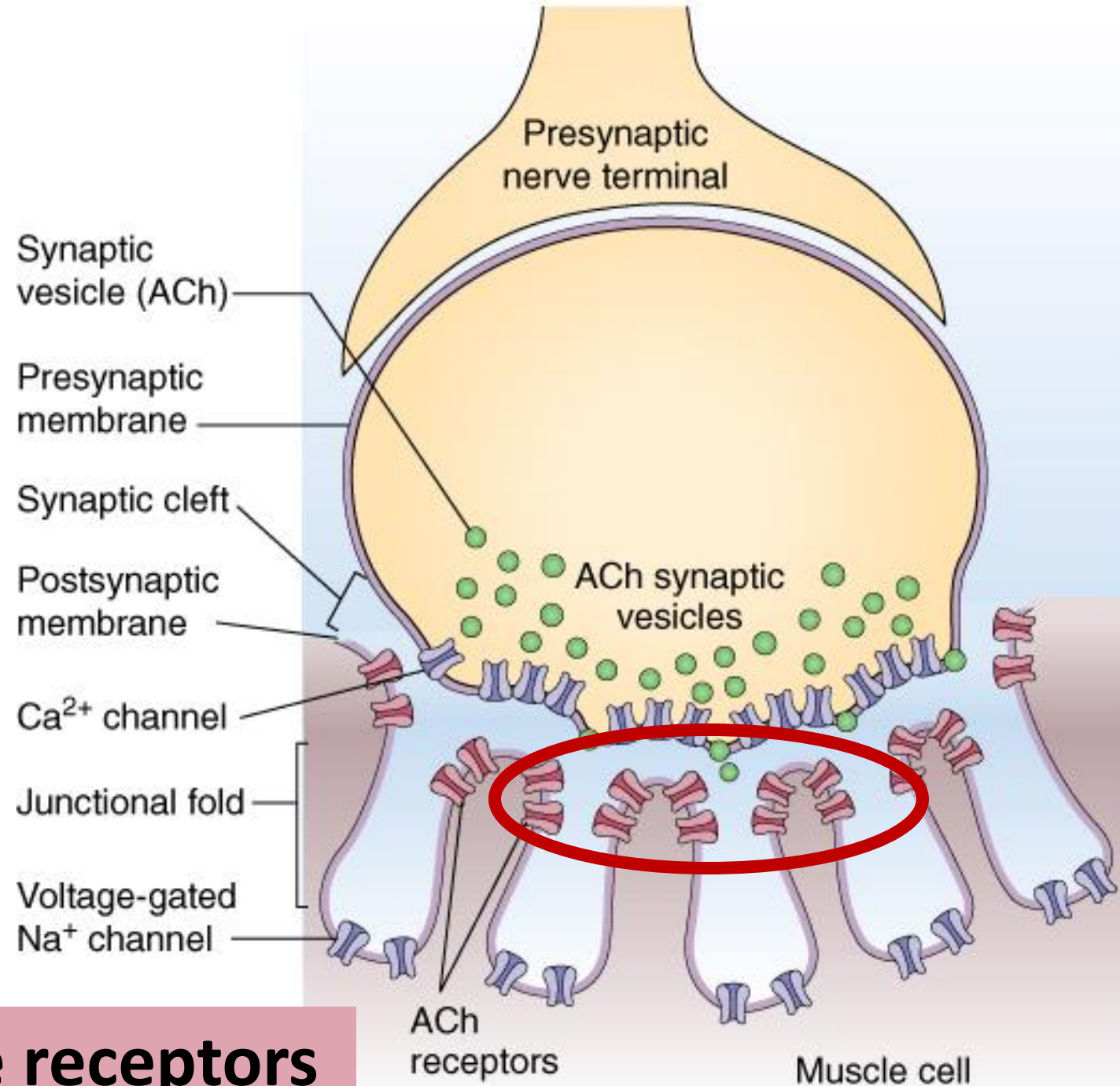
## 2. Gated ion channel: Ligand or Voltage-gated ion channel

- Ligand gated
  - Binding of neurotransmitter causes channel to open
  - Results in influx of ions through ion channel
  - Alters membrane potential → Promotes or inhibits nerve impulse transmission
- Some Examples:
  - **Nicotinic ACh receptors** in parasympathetic ganglia and Neuromuscular junction (muscle)
  - **Glutamate receptors (Ionotropic)** in CNS
  - **GABA-A ( $\gamma$ -aminobutyric acid) receptors** in CNS
  - **Glycine receptors** in CNS



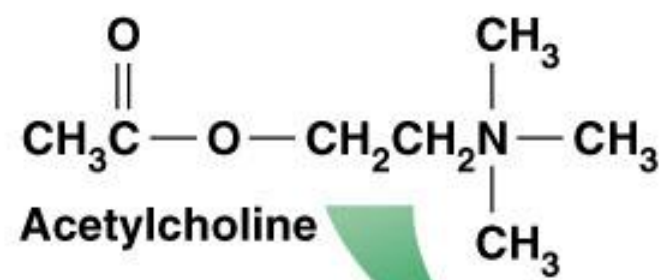
## 2. Gated ion channel

- Parasympathetic ganglia (Autonomic nervous system)
- Neuromuscular junction (muscle)

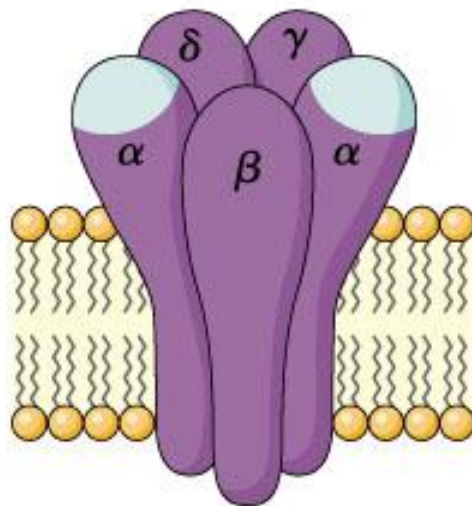


**Nicotinic Acetyl Choline receptors**

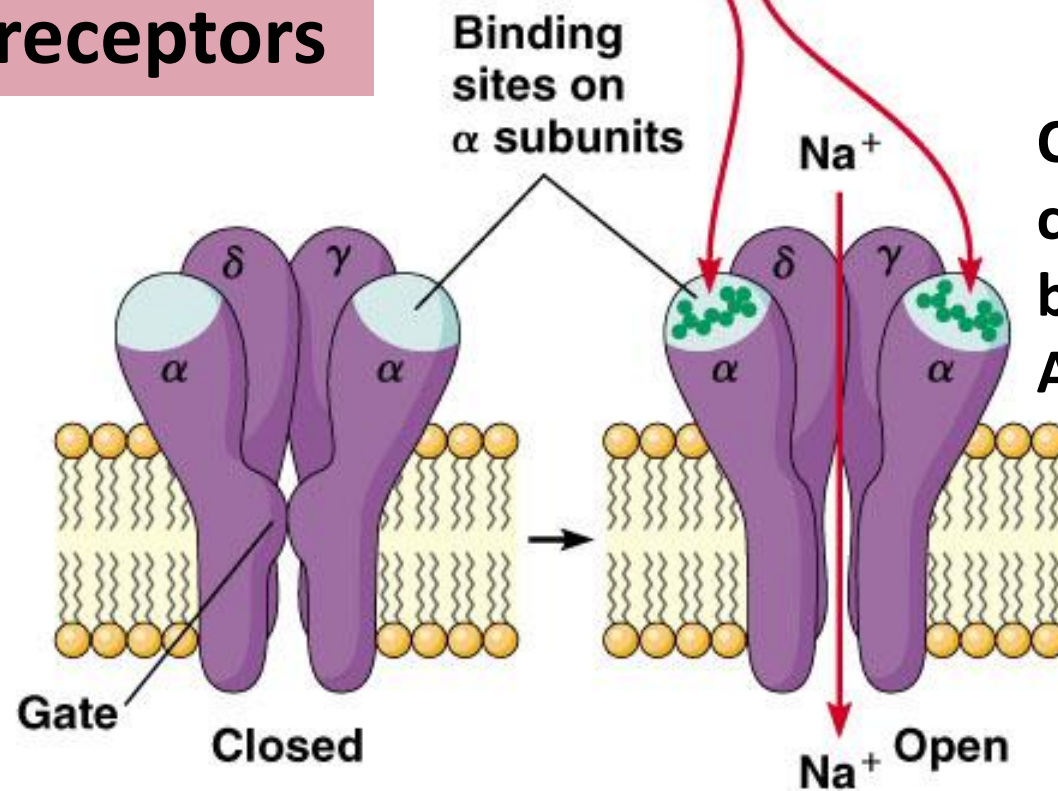
## 2. Gated ion channel



### Nicotinic ACh receptors



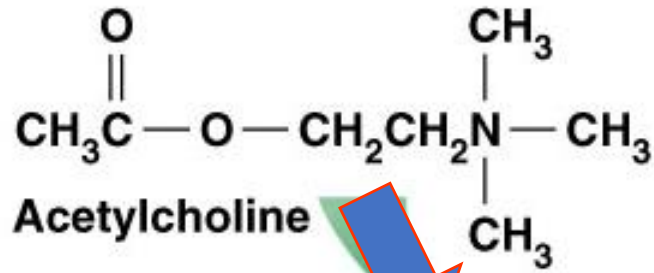
**(b) Structure of receptor**



**Conformational change on binding to Acetyl Choline**

**(c) Function of receptor**

# Termination of Acetylcholine signal

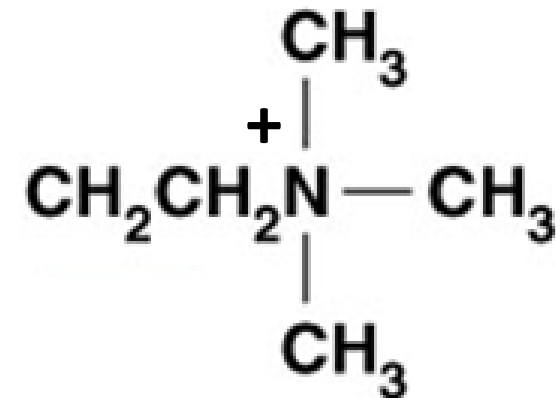
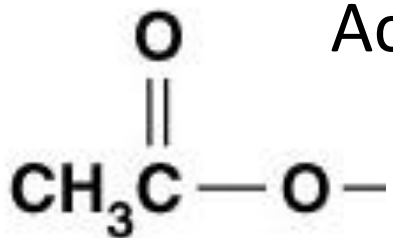


Enters synaptic cleft or neuromuscular junction

Signal is given!

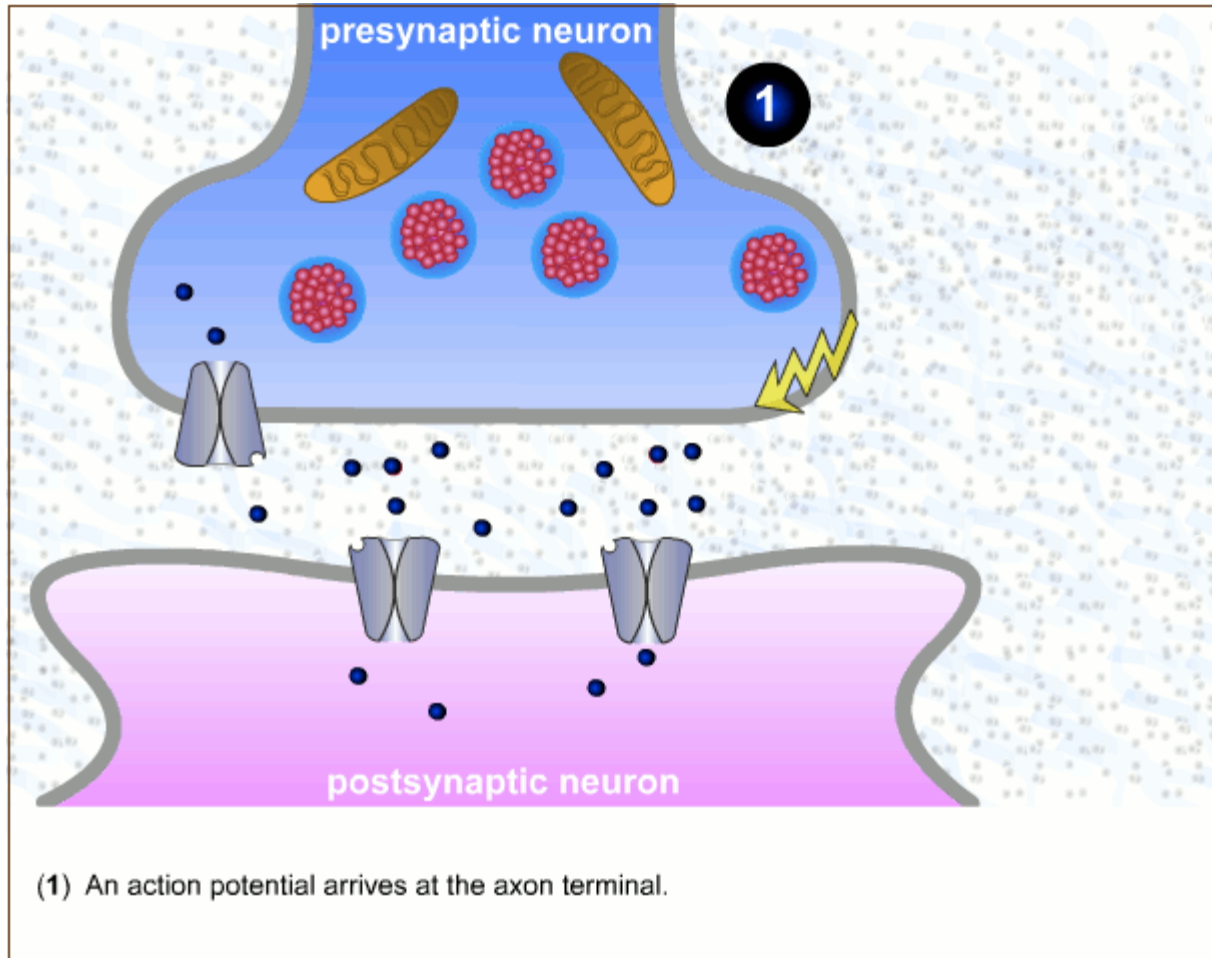
Cleaved by **Acetylcholine esterase** (an enzyme)

Acetate and choline may be recycled

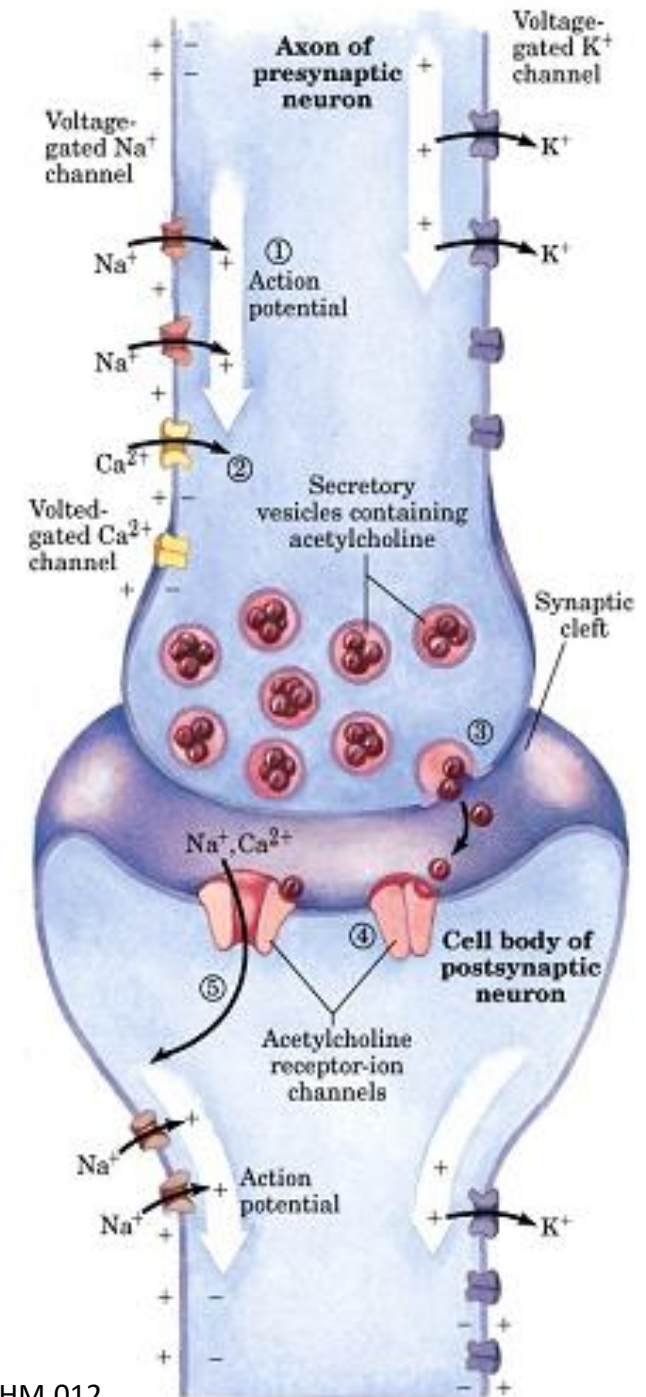




## 2. Voltage Gated ion channels respond to change in voltage



## Nicotinic Acetyl Choline receptors

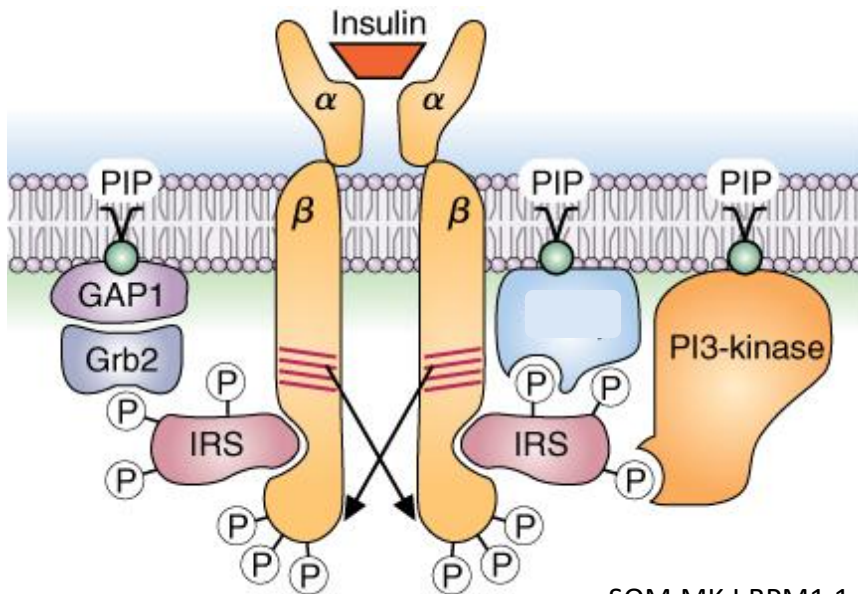


4 types of **signal transduction pathways** mediated  
by **4 different types of receptors**:

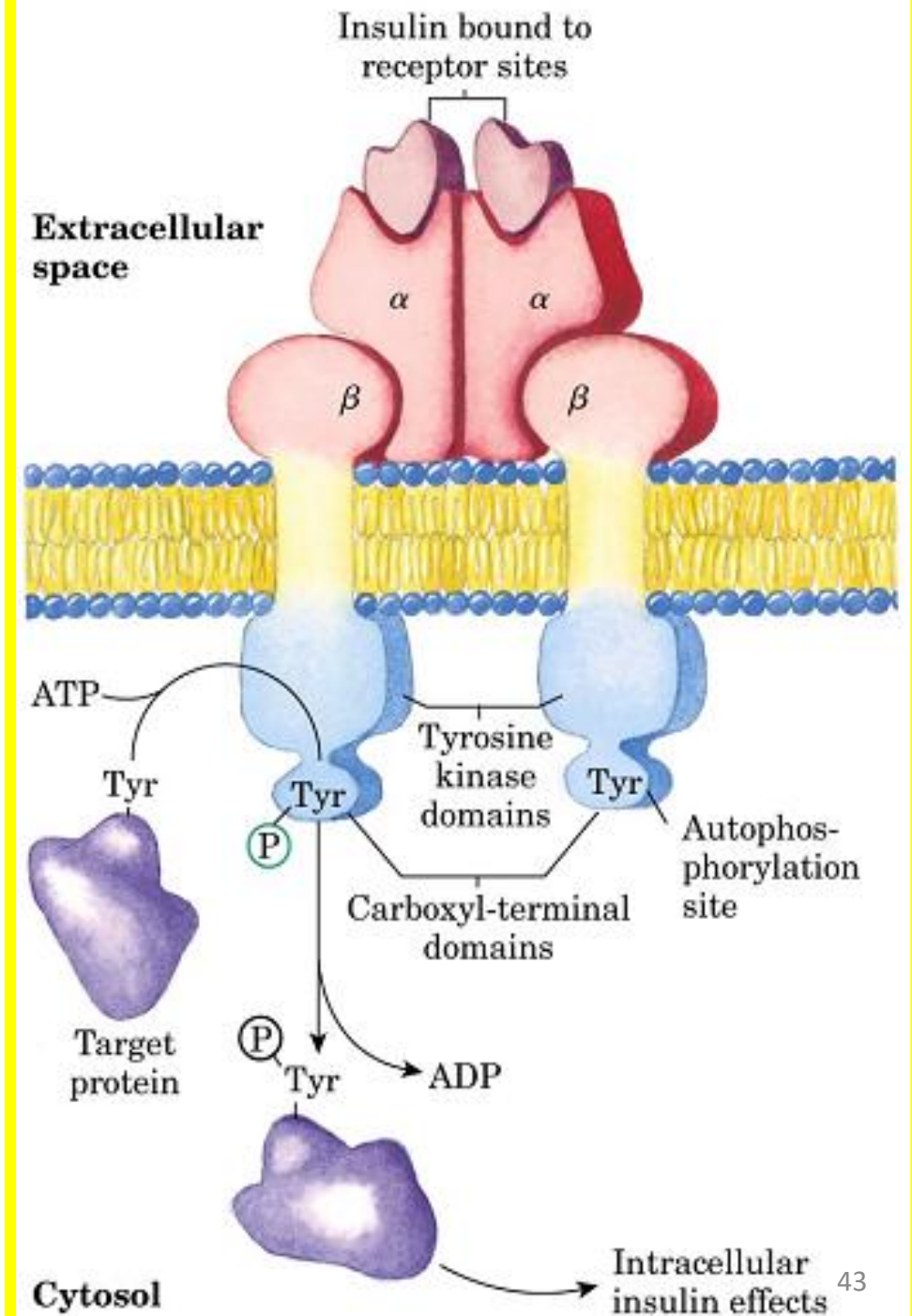
1. Steroid receptor
2. Gated-ion channel
3. Receptor enzyme (Catalytic receptor)
4. Receptors that form intracellular **second messengers**

### 3. Receptor enzyme (Catalytic receptor)

## Insulin Receptor Tyrosine Kinase as an example



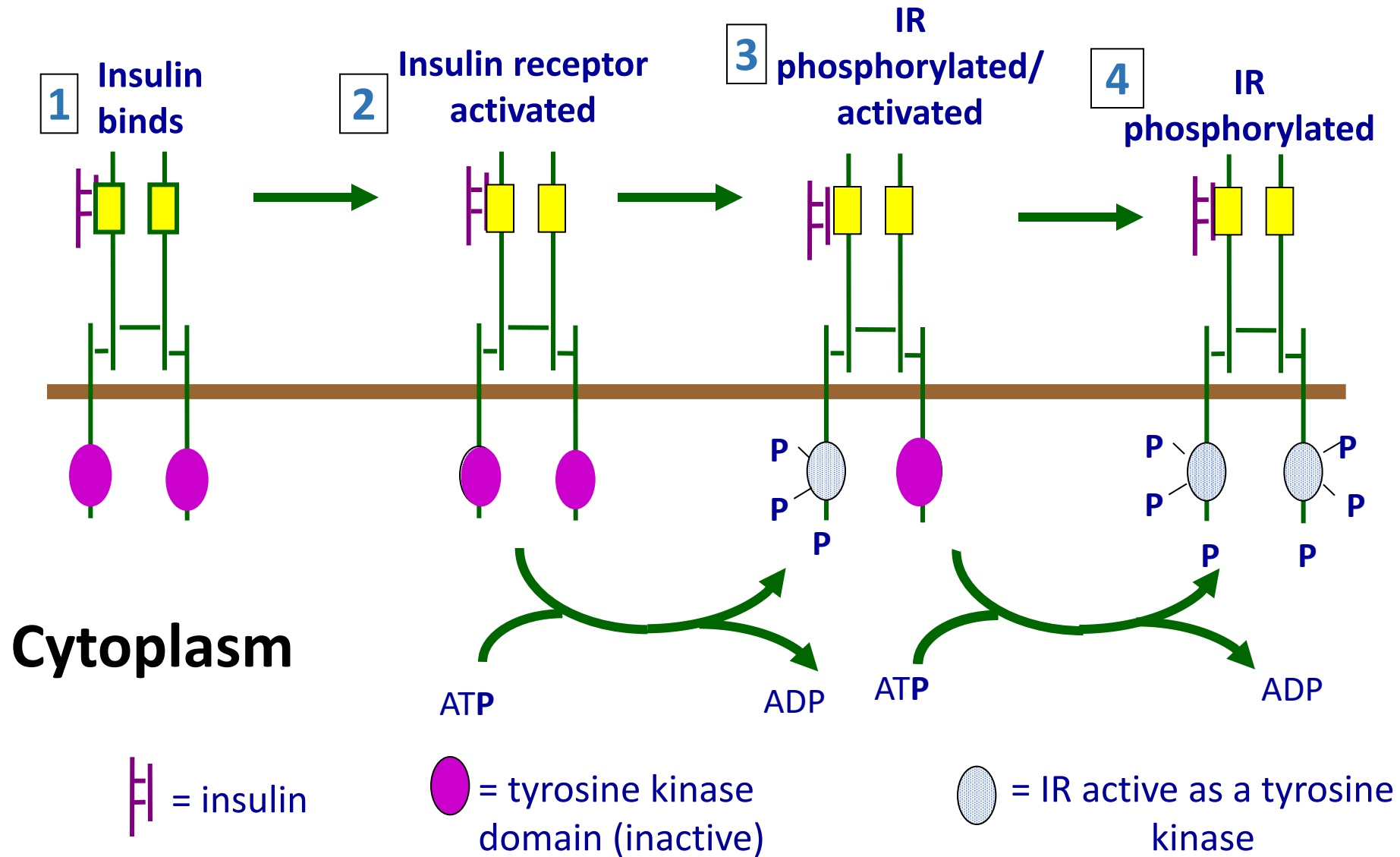
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### 3. Receptor enzyme (Catalytic receptor)

- Transmembrane catalytic receptors have intrinsic enzymatic activity
- Enzyme is **tyrosine-specific protein kinase (adds P to Tyrosine)**
- Insulin receptor
  - Extracellular domain for binding insulin and
  - Intracellular domain with **tyrosine kinase** activity
- Binding of ligand (Insulin) → ATP cleavage and **autophosphorylation** of tyrosine residues **Insulin receptor tyrosine kinase (IR) →** and then **phosphorylation of specific tyrosine side chains in target proteins**

# Extracellular



**Activation of tyrosine kinase domains of **IR** by insulin binding, followed by autophosphorylation**



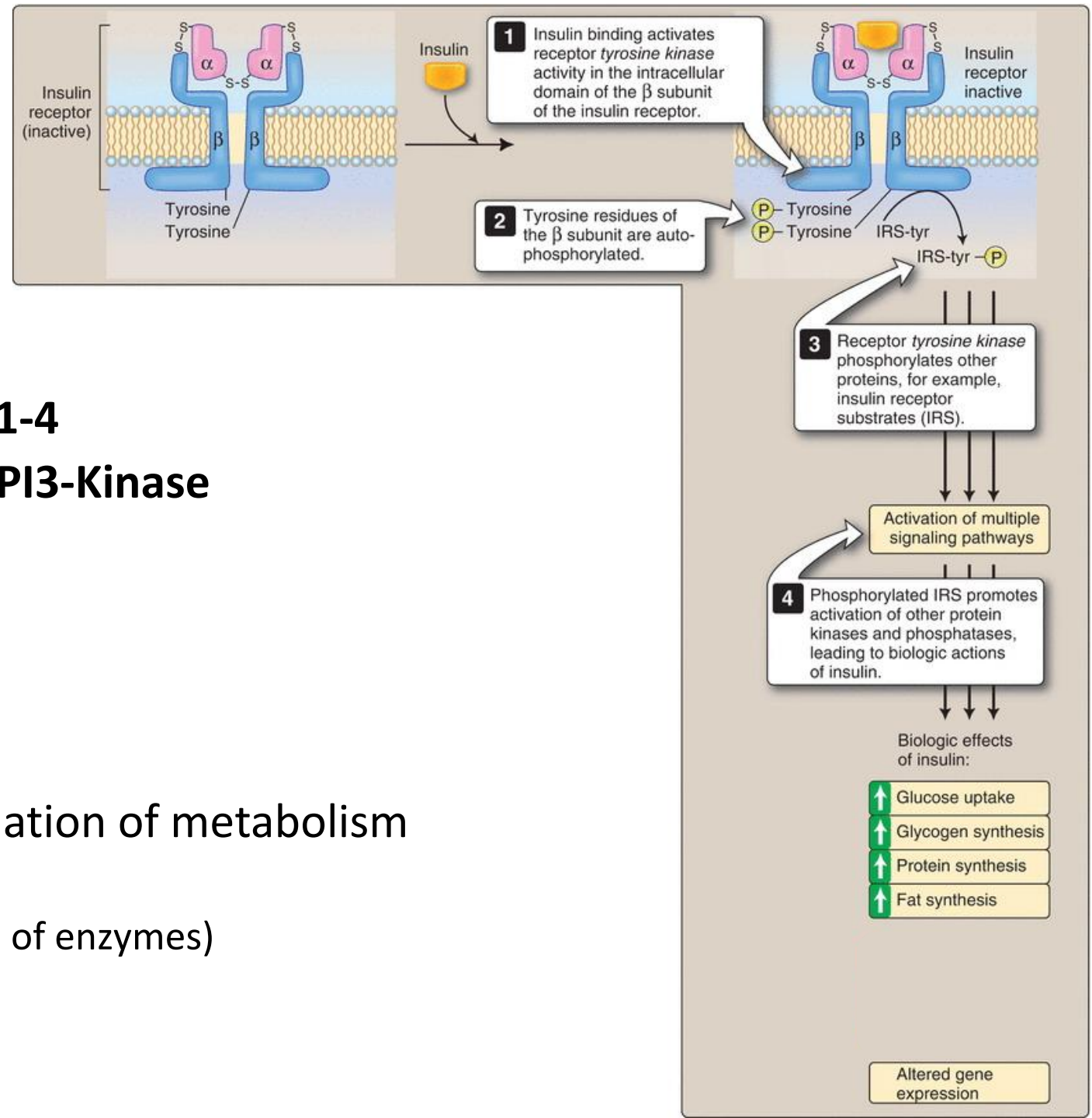
# Insulin Receptor: Intracellular effects

- Intracellular effectors:

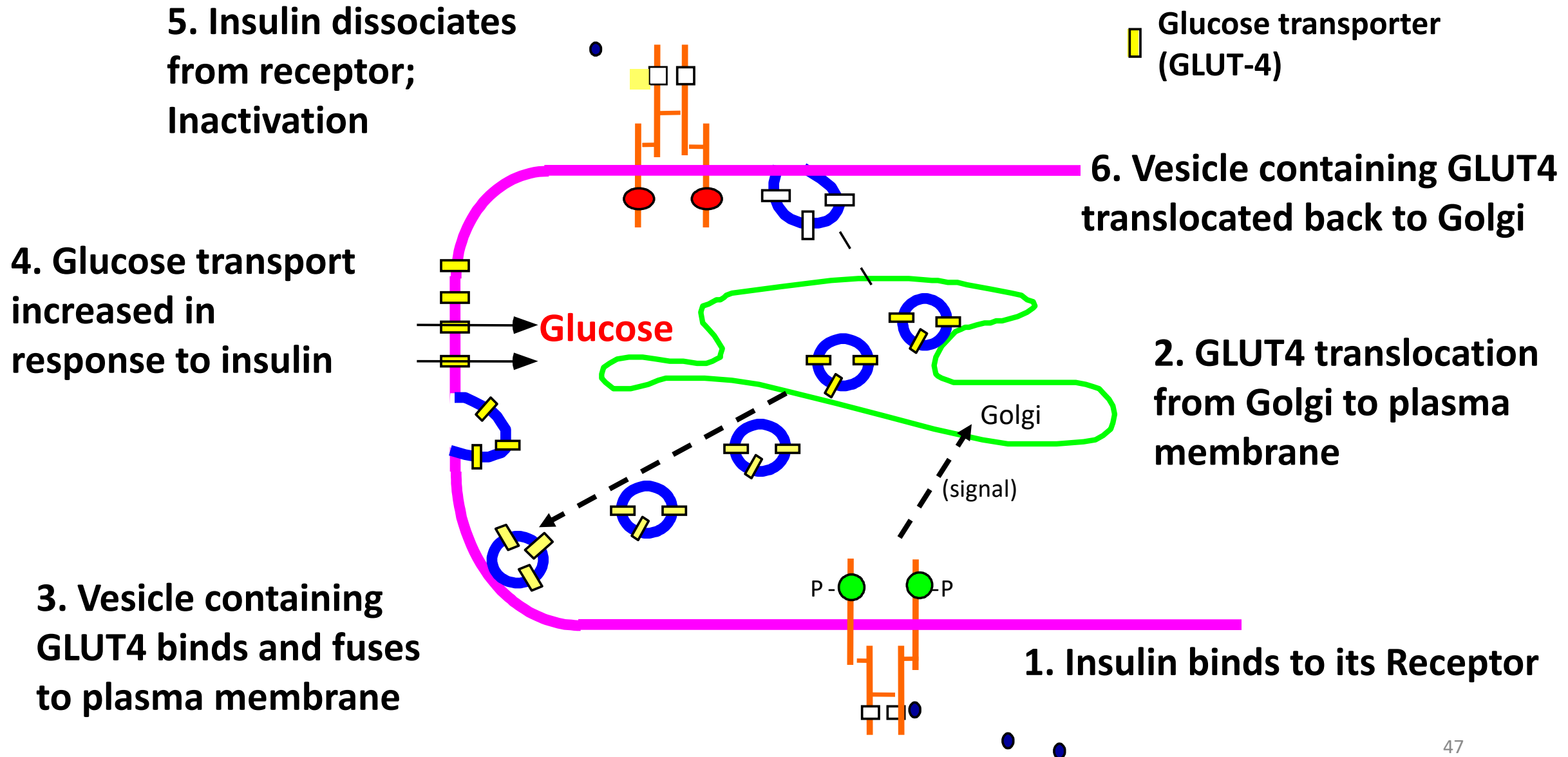
- Insulin receptor substrates: **IRS 1-4**
- Enzyme effectors: Activation of **PI3-Kinase**
- Multiple signaling pathways

- Cellular responses:

- Increased glucose uptake
- Regulation of transcription
- Activation of enzymes and regulation of metabolism
  - Covalent modification
  - Upregulation (increased synthesis of enzymes)



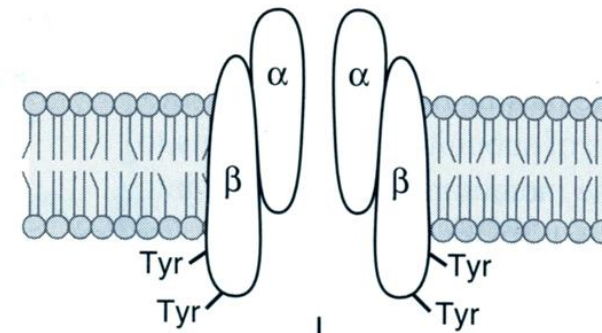
# Insulin stimulated glucose transport (GLUT-4) in adipose or muscle cells



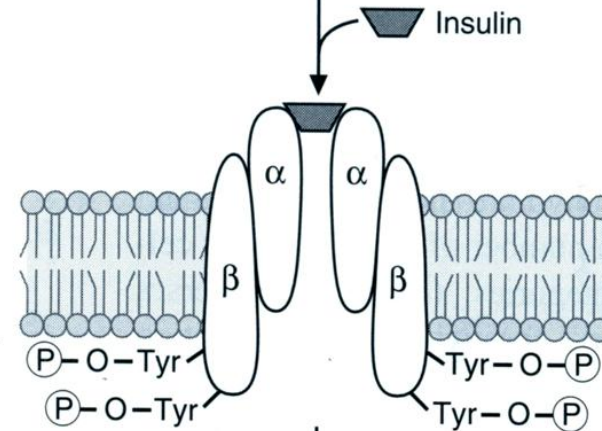


# Insulin Receptor: Intracellular effects

The insulin receptor is a dimer with two different types of subunits. It spans the membrane.



The  $\beta$ -subunits are tyrosine kinases. When insulin binds the subunits phosphorylate themselves at tyrosine residues.



The subunits also phosphorylate other protein kinases at tyrosine residues. These kinases produce the cellular responses.

Insulin resistance:  
Post-receptor defect

Reversal of  
glucagon-stimulated  
phosphorylation

Phosphorylation  
of proteins

Induction and  
repression of  
specific genes

Stimulation of  
general protein  
synthesis

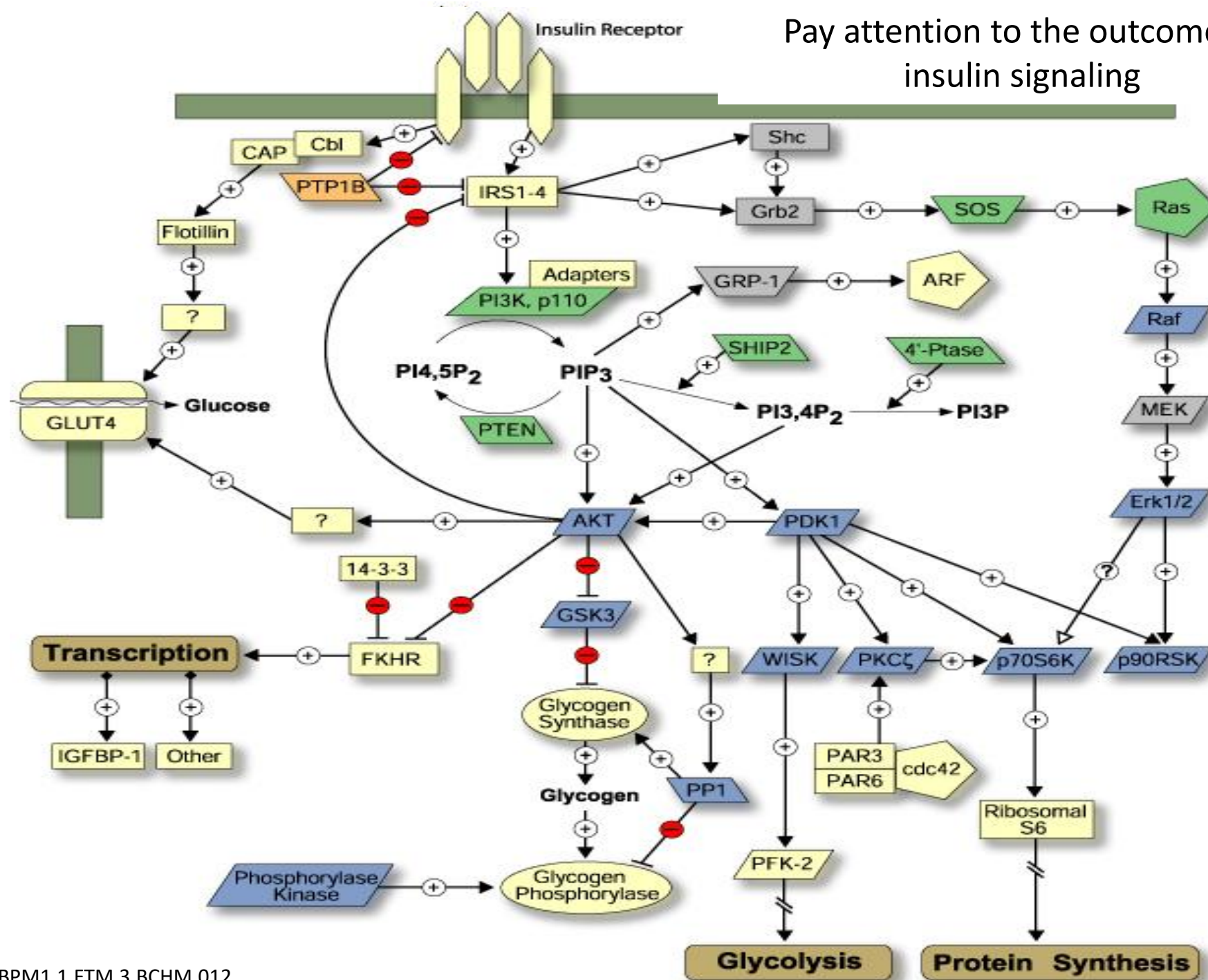
Stimulation of  
glucose transport

**GLUT4**

**Type 2 Diabetes mellitus**

**Fig. 24.17.** Insulin receptor initiates tyrosine phosphorylation and cellular responses to insulin.

Pay attention to the outcome of insulin signaling

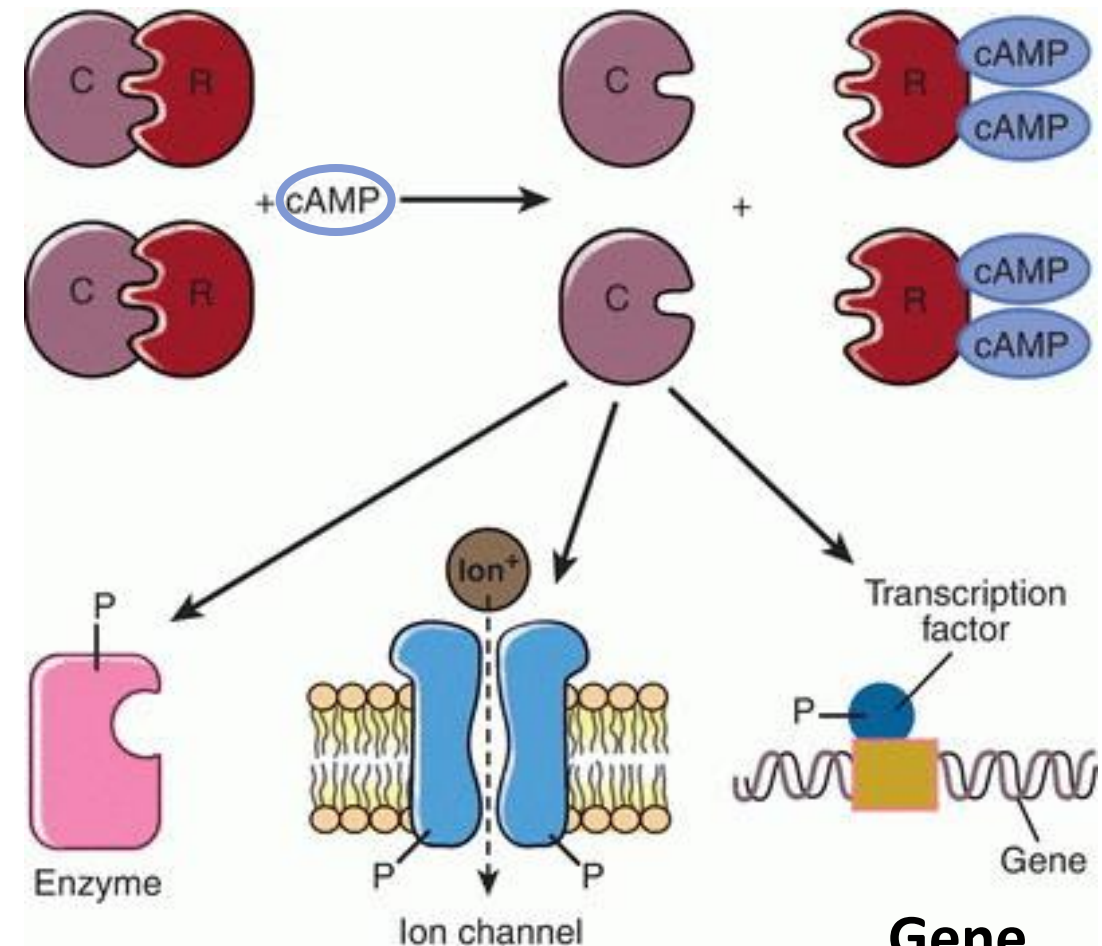


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4. Receptors that form intracellular **second messengers**
  - G-protein coupled receptor (GPCR)
  - Guanylate cyclase

# 4. Receptors that form second messengers

- **Receptors pass signal (message) from outside cell by producing second messenger inside the cell**
- Types of receptors
  - **G-protein coupled receptors**
  - Guanylate cyclase (cGMP) System
- Rapid and immediate effects
  - Phosphorylation of enzymes (activation or inactivation of key regulated enzymes)
  - Changes in ion levels (calcium)
- Slower effects on gene expression



**Covalent modification  
of enzymes**

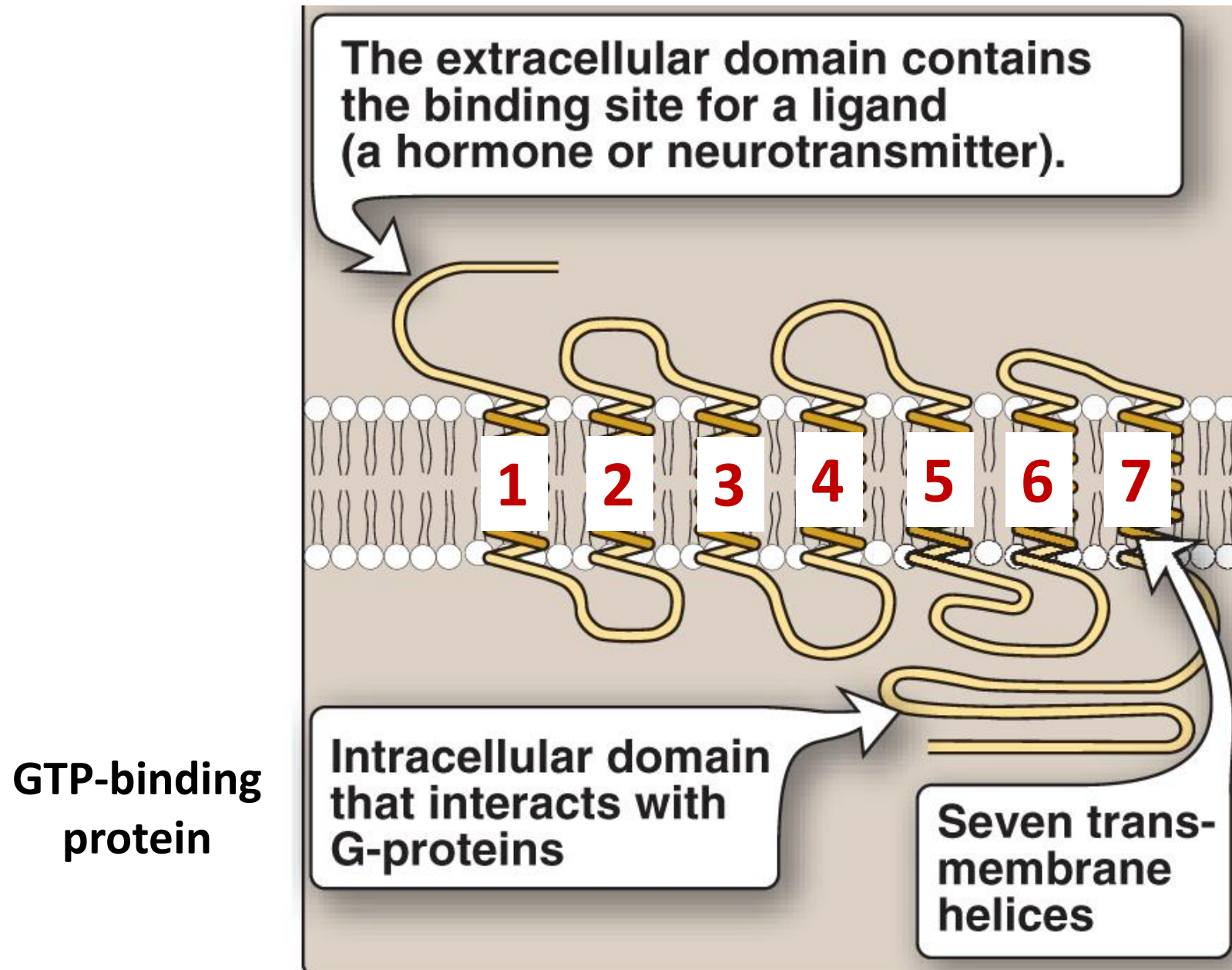
**Gene  
Induction/  
Repression**

## 4. G-protein coupled receptors and intracellular second messenger

- Hormone (first messenger) changes concentration of some other molecule (second messenger) inside the cell
- Generally **small, non-protein molecules**
- Produced **intracellularly** when hormone binds to its receptor
- Initiate cascade of intracellular events (enzyme activation/inhibition, altered gene regulation) resulting in specific cellular responses
- Examples:
  - Adenylate cyclase system produces **cAMP**
  - Calcium - Phosphatidylinositol system produces **IP3, DAG, Ca<sup>2++</sup>**



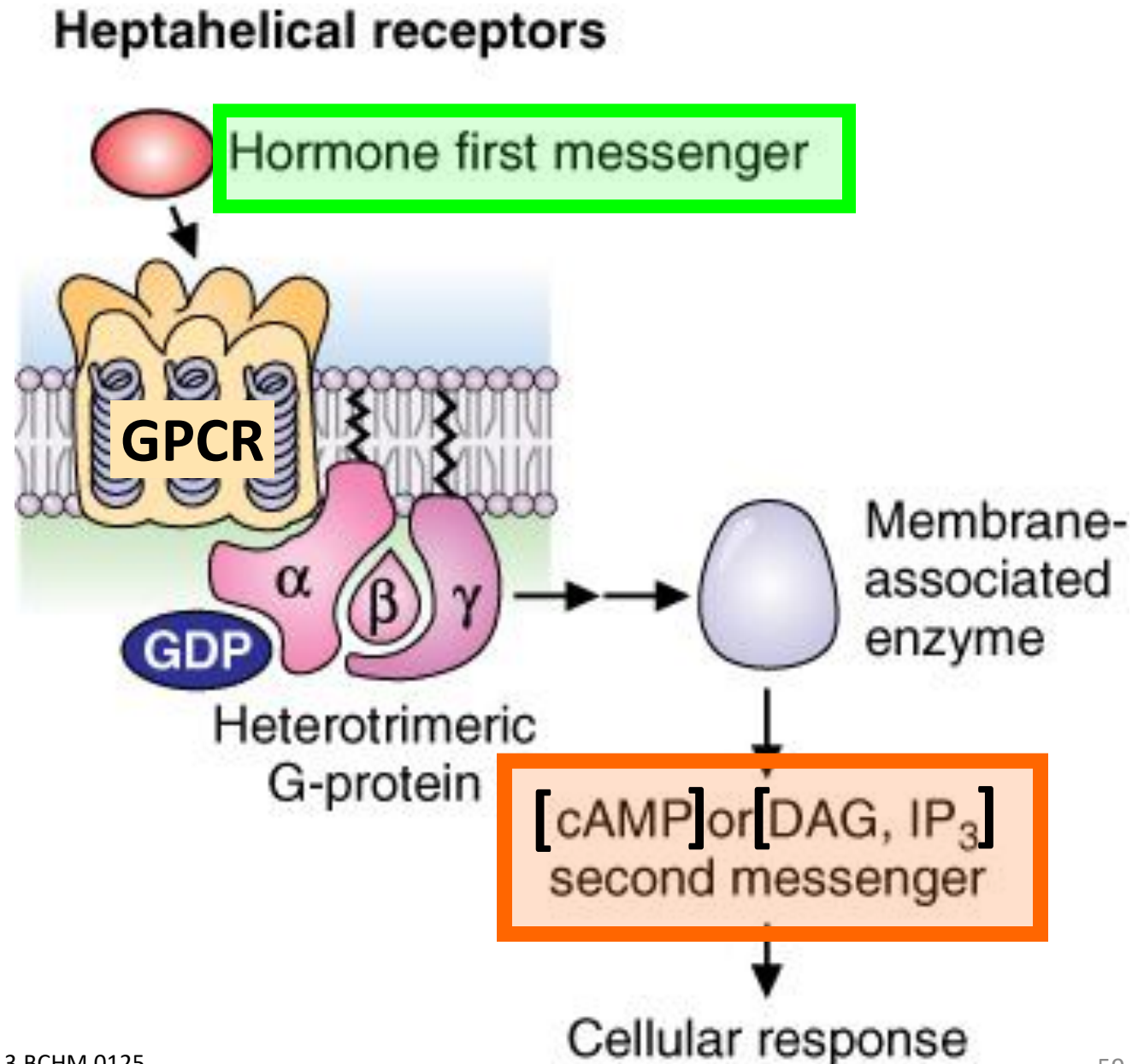
#### 4. Heptahelical receptors, G-protein coupled receptors (GPCR's) and second messengers



#### 4. Heptahelical receptors (GPCR's) and **second messengers**

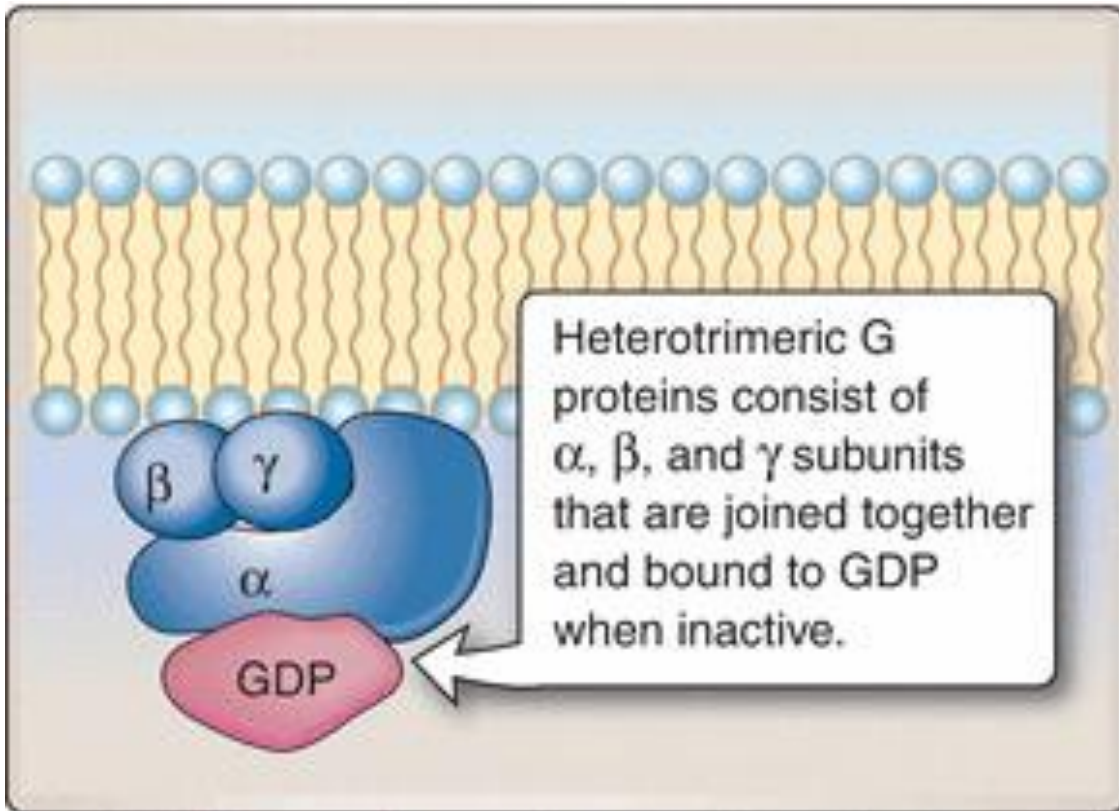
- Hormone (or cytokine or neurotransmitter) is **first messenger**
- Binds plasma membrane heptahelical receptors

- **Hormone–receptor complex** activates **G-protein** that activates **membrane-bound enzyme (effector enzyme)**
- Activation of different GPCRs lead to generation of different intracellular second messengers, such as cAMP or DAG/IP3.





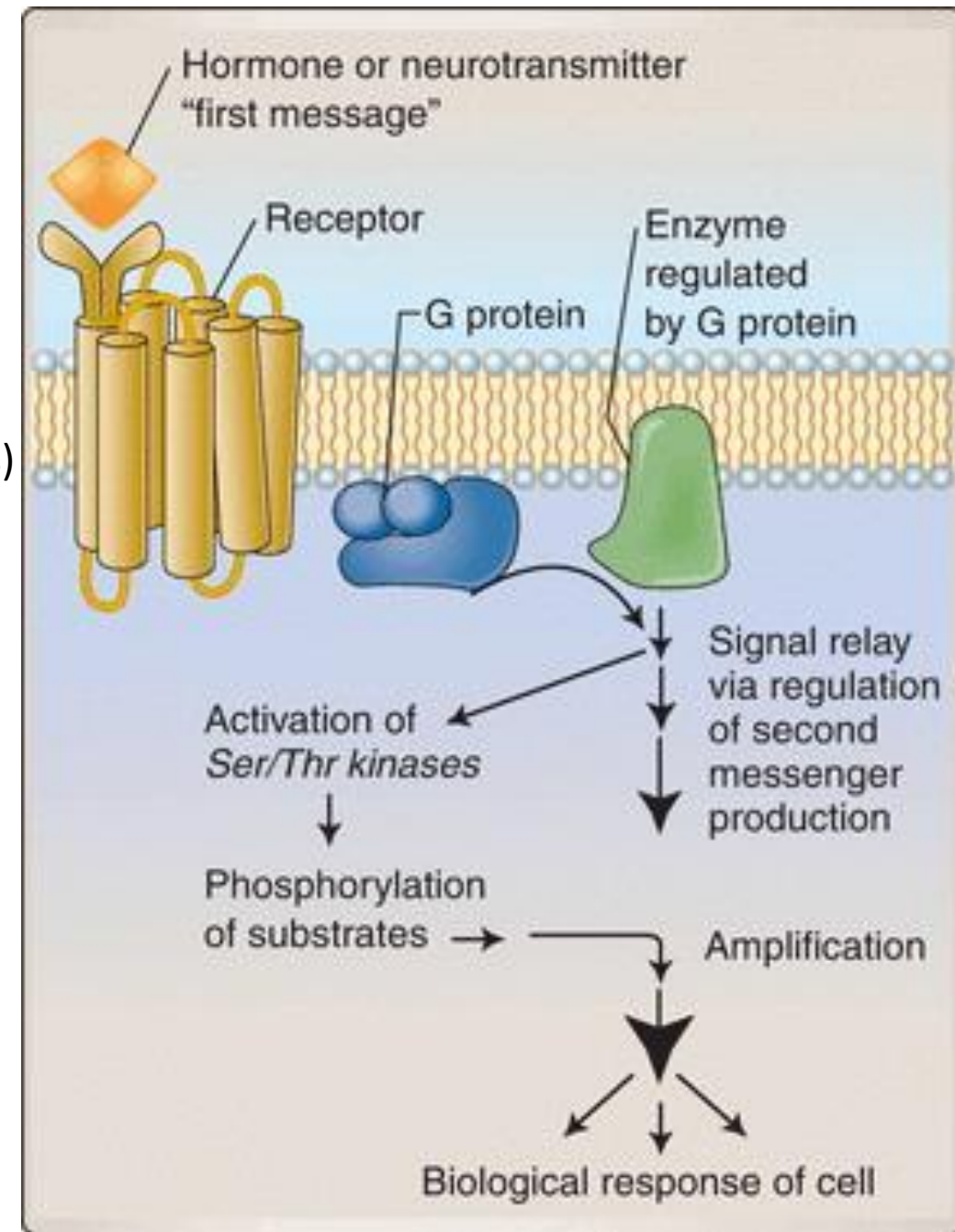
# Heterotrimeric G proteins



- Binding of hormone to receptor activates G protein
- The  $G\alpha$ -subunit dissociates; binds GTP ( **$G\alpha$ -GTP**) is active
- $G\alpha$ -GTP activates **enzymes** to produce intracellular second messengers
- $G\alpha$  subunit has intrinsic GTPase activity (**intrinsic timer**) – termination of signal

#### 4. Heptahelical receptors (GPCR's) and second messengers: Types of $G\alpha$

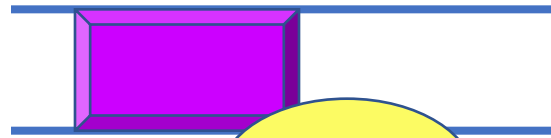
- **Alpha1 adrenergic receptor:  $G\alpha_q$** 
  - Activates **phospholipase C** (enzyme)
  - Increases **IP3, DAG and  $Ca^{2+}$**  (second messengers)
- **Alpha2 adrenergic receptor (inhibitory):  $G\alpha_i$** 
  - **Inhibits adenylate cyclase**
  - **Decreases cAMP levels**
- **Beta-adrenergic receptor; Glucagon (stimulatory):  $G\alpha_s$** 
  - **Activates adenylate cyclase**
  - **Increases cAMP levels**



Adrenergic receptors  
generate different  
second messengers

## Epinephrine or Norepinephrine

$\alpha$ -1 adrenergic receptor



Phospholipase C

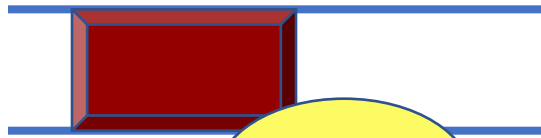
Phosphatidylinositol  
bisphosphate (PIP2)

**IP3**

**DAG**

**Ca<sup>2+</sup>**

$\alpha$ -2 adrenergic receptor

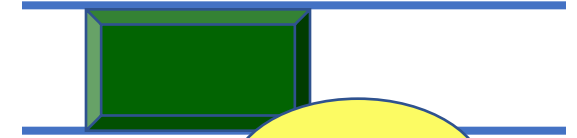


Adenylate cyclase

ATP

**↓cAMP**

$\beta$ -adrenergic receptor



Adenylate cyclase

ATP

**↑cAMP**